

ROBOTICS 101

Introduction To Robotics

NSM ROBOTICS

L 1

Welcome To This
Course on
Introduction To Robotics (Robotics 101)
↳ ROBOTICS 101

About this Course :

3D

- ① It will cover the Robotics subject holistically. ↑
- ② Balanced approach → Theory, Numericals etc. Practical
- ③ Simplified Concept using 3D CAD and Simulation softwares for better understanding.

Robotics 101
102
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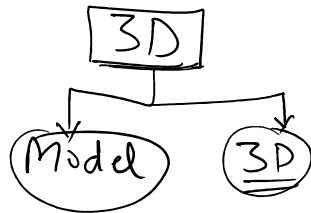
Language — Hindi (+ English) ✓
Course Material — English. ✓

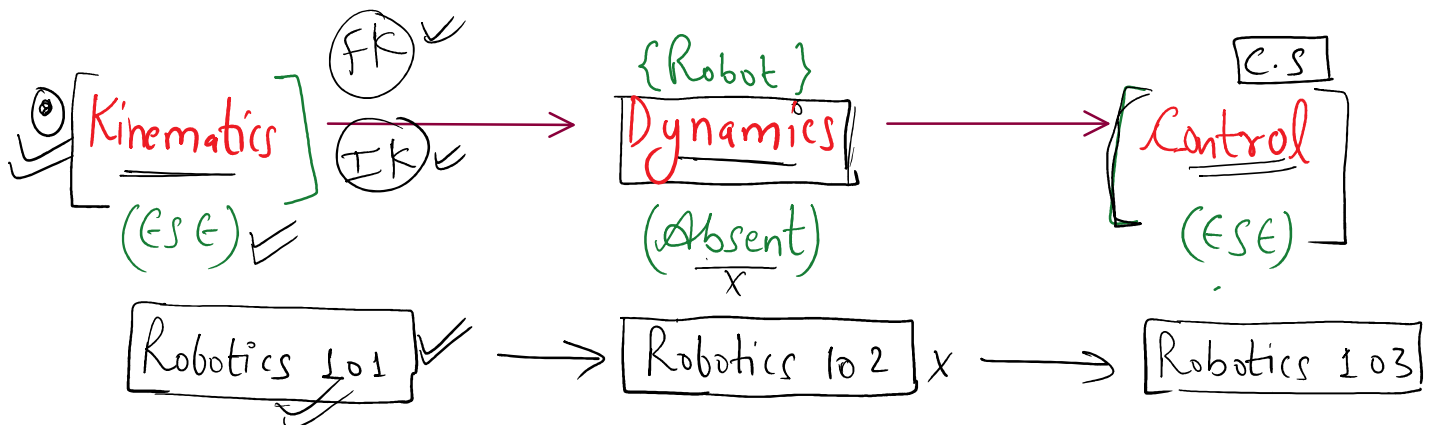
Is it useful for exams?

Engineering Services Examination (ESE) — ME
Mechanical Engineering (ME)
Subject : Robotics

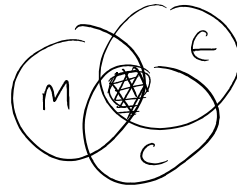
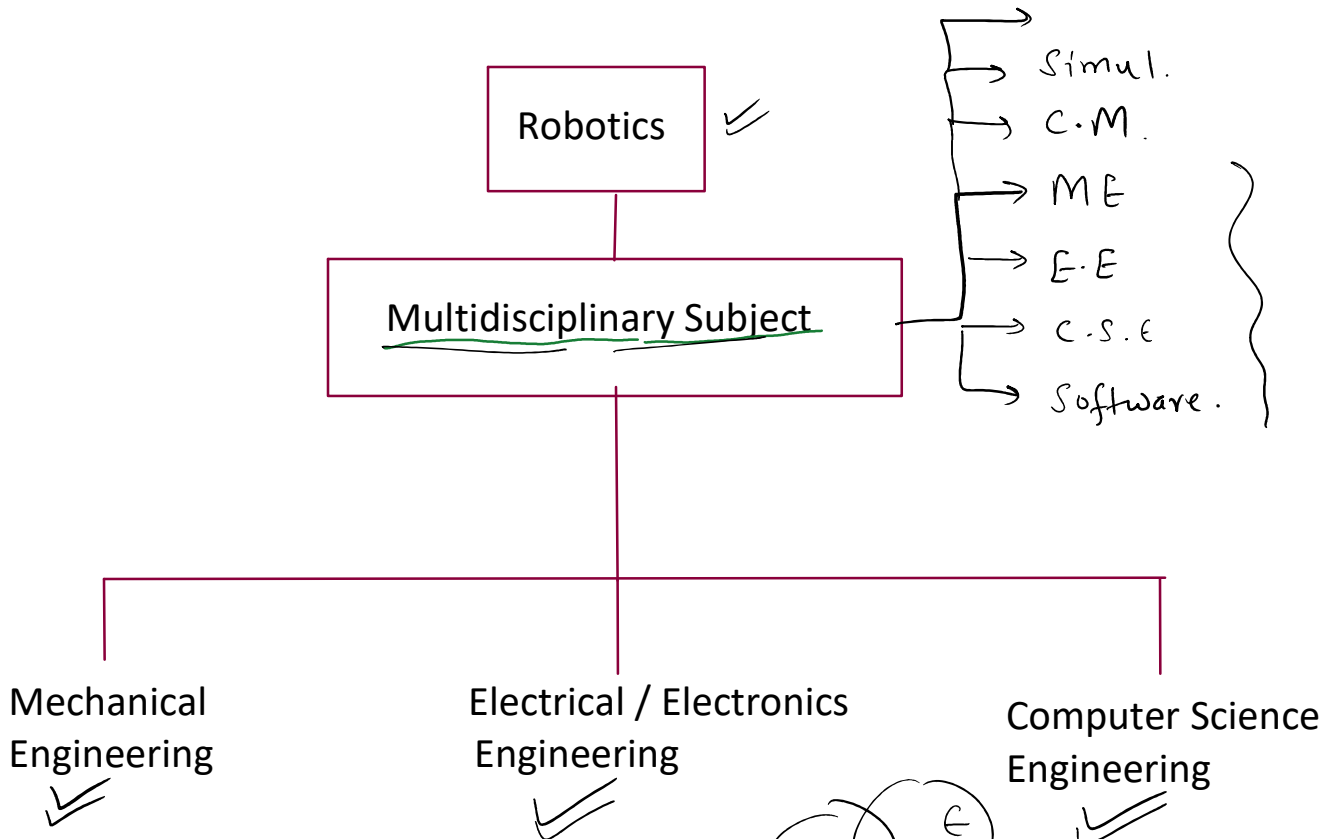
R + M + C

Syllabus

- ① Robot Classification → Types $\begin{cases} \text{(i)} \\ \text{(ii)} \\ \text{(iii)} \end{cases}$
 - ② Robot Specification
 - ③ Notations
 - ④ Kinematics $\begin{cases} \text{Direct} \\ \text{Inverse} \end{cases}$  R 101
 - ⑤ Homogenous Transformation coordinates
 - ⑥ Arm equations of four axis SCARA robot
- ←—————→



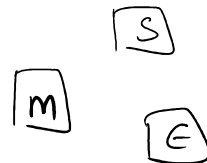
Subject - Nature



○ To be a true robotist is very difficult

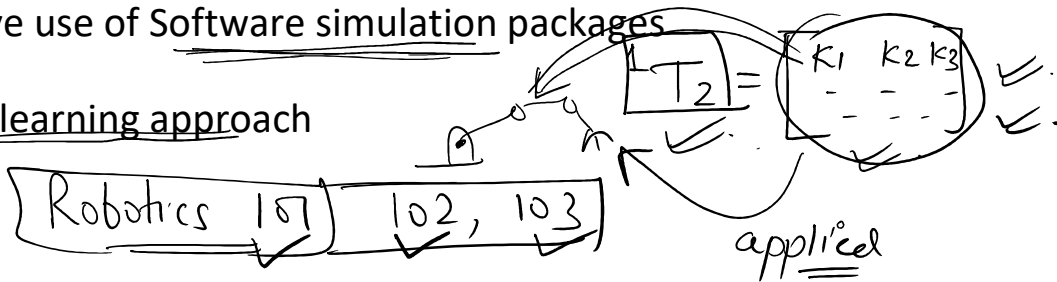


○ works in teams of specialized skills / Disciplines



○ Extensive use of Software simulation packages

○ Holistic learning approach



So we will learn this course from a mechanical engineer's point of view and will focus primarily on **ESE EXAM SYLLABUS**

Robot — Definition

What is a Robot ?

o Multiple definitions exists, no one is the Standard definition.

O variation across different Robotic associations

O variation across different (applications)

O for academic purpose, we will study some accepted and frequently used definitions

So, how do we basically define a Robot ?

lets study few definitions

1) International standards organization (ISO)

2) Robotics Industries Association (RIA)

3) British Robot Association

Definition

British →

"A reprogrammable device with minimum of four degrees of freedom designed to both manipulate and transport parts, tools, or specialized manufacturing implements through variable programmed motions for performance of specific manufacturing task."

ISO →

"An industrial robot is an automatic, servo-controlled, freely programmable, multipurpose manipulator, with several areas, for the handling of work pieces, tools, or special devices. Variably programmed operations make the execution of a multiplicity of tasks possible."

American (RIA)

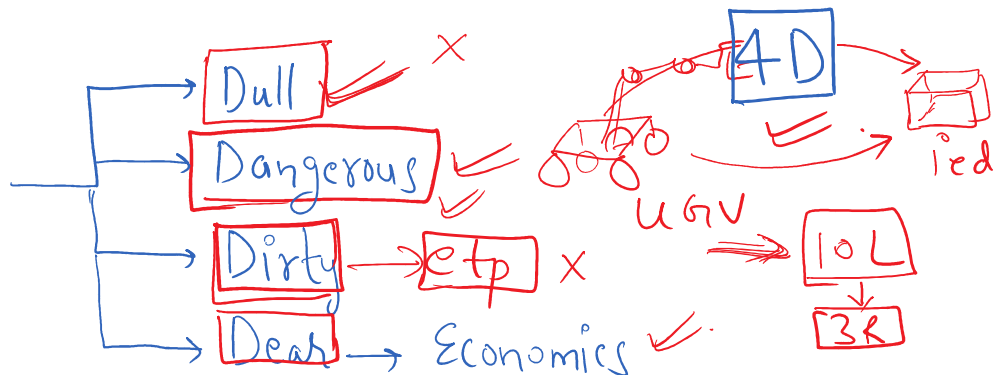
"A reprogrammable, multifunctional manipulator designed to move material through variable programmed motions for the performance of a variety of tasks."

- ⇒ Automatic ✓
- ⇒ Reprogrammable ✓
- ⇒ Multifunctional + Variably programmed ⇒
- ⇒ Manipulator / Manipulation 

To put simply →

A robot can be defined as a machine that can replace human beings as regard to physical work and decision making.

Robots are preferred where



Laws of Robotics

Three Laws of Robotics

The **Three Laws of Robotics** (often known as Asimov's Laws) are a set of rules devised by science fiction author **Isaac Asimov**. The rules were introduced in his **1942** short story "**Runaround**" (included in the 1950 collection I, Robot), although they had been foreshadowed in some earlier stories. The Three Laws, quoted from the "Handbook of Robotics, 56th Edition, 2058 A.D.

Credit and Source : Wikipedia

First Law

A robot may not injure a **human being** or, through **inaction**, allow a human being to come to harm.

Second Law

A robot must obey the orders given it by **human beings** except where such orders would conflict with the **First Law**.

Third Law

A robot must protect its own existence as long as such **protection** does not conflict with the **First or Second Law**.

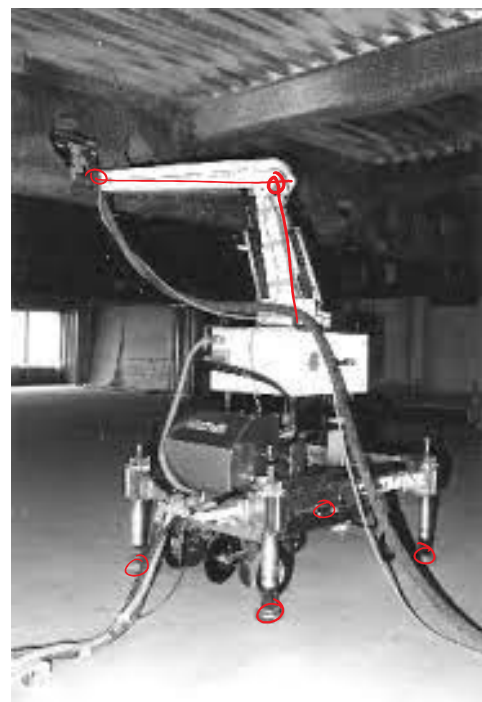
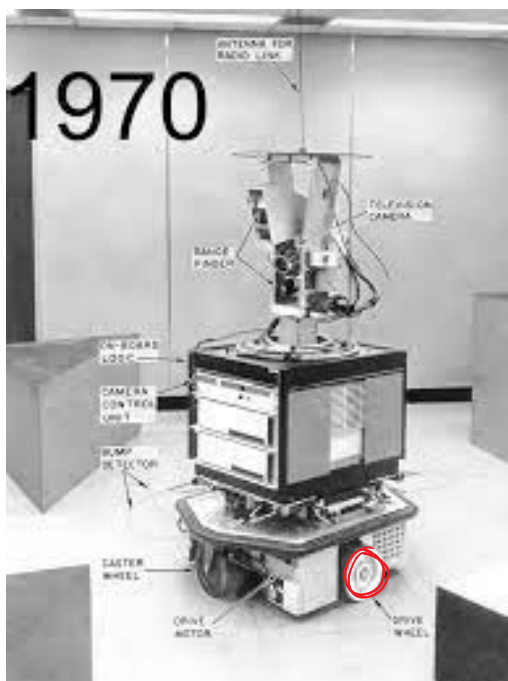
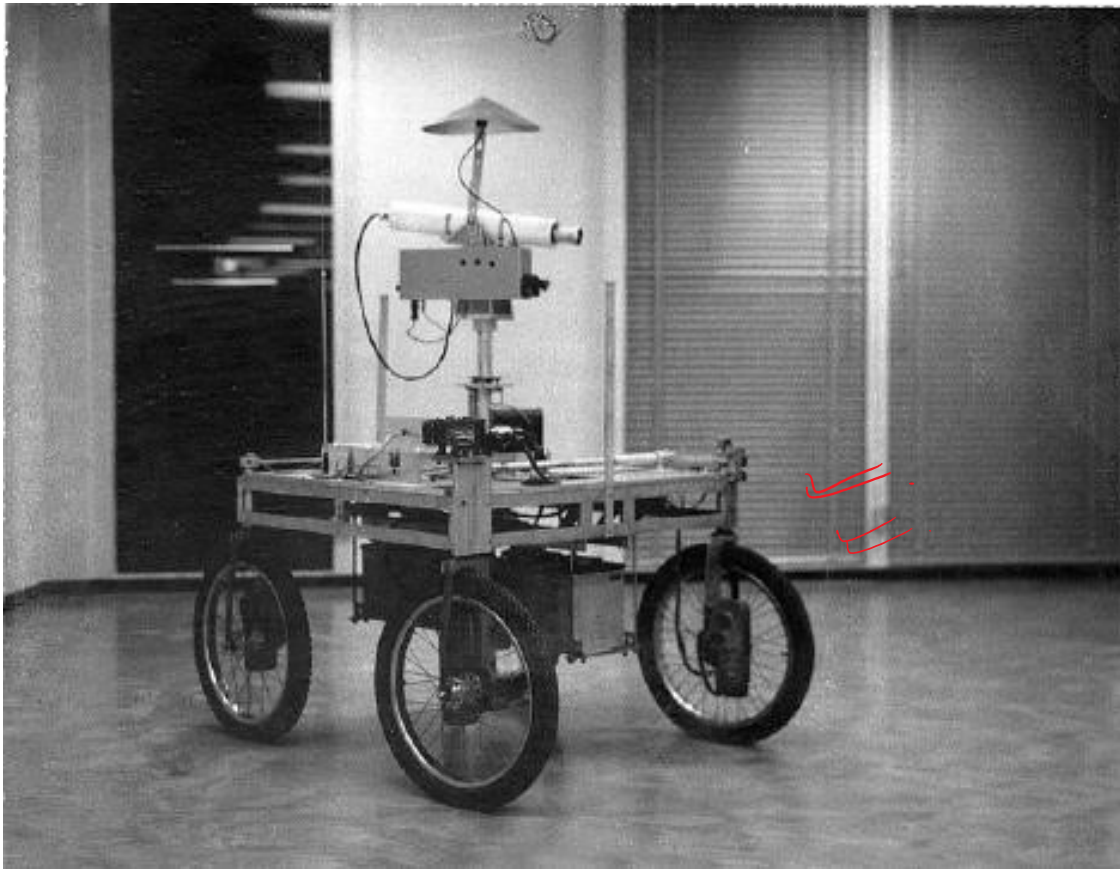
later Several other laws we added

for example, **Asimov's fourth law**

Asimov's fourth law :

A robot may take a human being's job but it may not leave that person jobless.

Robotic Systems (at the beginning of Industry)



Modern Robotic Systems

Types of Robots

Humansoids



Legged Robots



Self Driving Cars



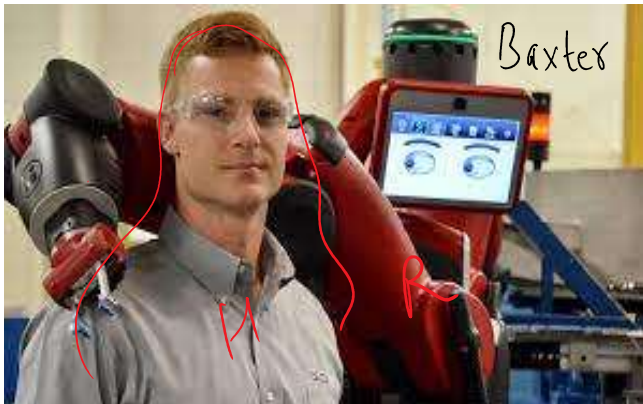
Military Robots



"Bio" Inspired / Bionic Systems



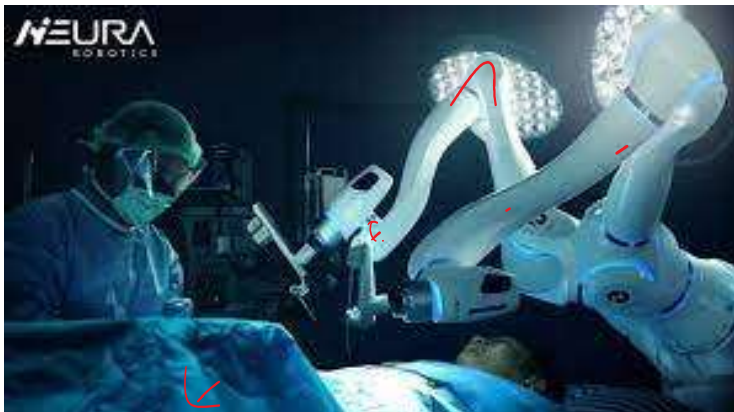
Collaborative robots (Cobots)



Aerial Robots



Medical Robots



Advantages of Robotic Systems

- 1) Safety ✓✓
- 2) Speed ✓✓
- 3) Accuracy ✓✓
- 4) Repeatability ✓✓
- 5) Consistency ✓✓
- 6) Perfection ✓✓
E }
- 7) Productivity ✓✓ 8h → 2p | 8h → 5p.
- 8) Efficiency ✓✓
- 9) Employee's workload ↓ ✓✓
50 10h SR
8h
- 10) Job / Industry Creation ✓✓
- 11) Robotic Technology use in "Other fields"
P
C
A
RP
- 12) "Multi Skills" → M, E, C.S, IT

Disadvantages of Robotic Systems

- 1) Economic stress ↑ on Company 100 Cr.
- 2) Job loss / Unemployment / Profession change x
- 3) Lacks human intelligence / "emotions" Ex scale
- 4) Less Dextrous
- 5) Very Costly / needs very skilled operators
- 6) High maintenance and power needs ↑
- 7) Concern about AI and its "disruptive" capabilities
- 8) Increased military use ↑ Biased ;
→ Dual use Tech. USA

Use of Robotic Systems

Manufacturing ✓

Working alongside humans in manufacturing plants
(co-bots)

Assembly ✓

Inspection ✓

Packing ✓

Transport ✓

Earth and Space exploration } NASA MARS ROVER.

Nuclear Installations

Medicine and Surgery ✓

Defence and Weaponry

Academics and Research

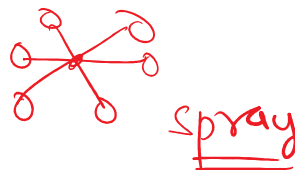
Companions, Waiters, Assistants, Vacuum Bots etc.

Search-and-rescue missions

Logistics Chain - Amazon warehouses - Self Driving Bots → UAV

News Readers, Teaching Assistants etc.

Farming and Agriculture ✓



Armed Forces / Military.

Underwater Robots, Aerial Robots, Soft Robots

Controversy / Disputes

→ (Self-crashing cars)

→ Political manipulation bots/ computational propaganda

→ Algorithms for peace (["]usaf ["]project maven)

→ surveillance

→ Deep fake

→ Algorithmic discrimination

Ai overtake - exascale computation

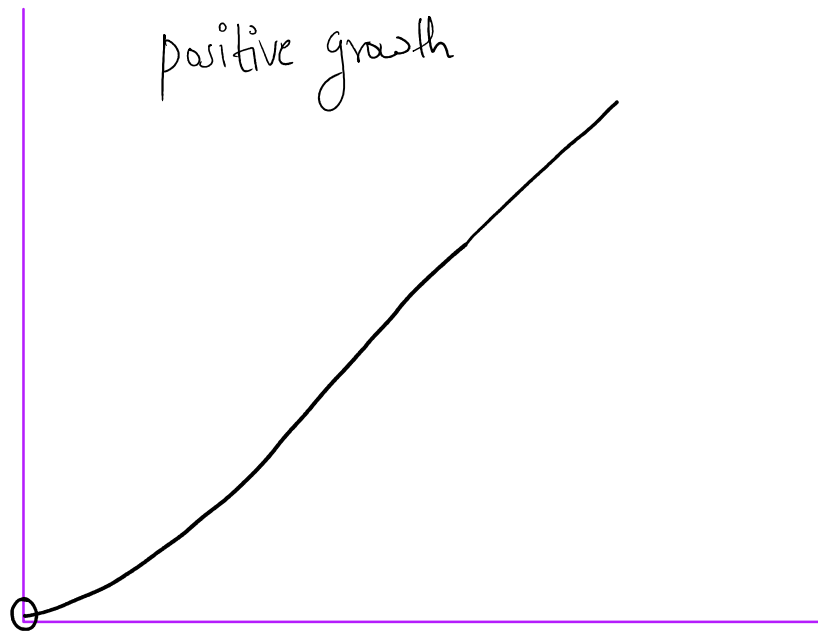
Credit : MIT technology review

Robotics and Mechatronics in ESE and other exams?

- (i) India $\rightarrow$ Manufacturing Hub $\rightarrow$ 10 Tr - 12 Tr USD by 2030
- (ii) Startups and Innovations.

NEXT DISRUPTION

The global robotics technology market size was valued at \$62.75 billion in 2019, and is projected to reach **\$189.36 billion by 2027**, growing at a CAGR of 13.5% from 2020 to 2027



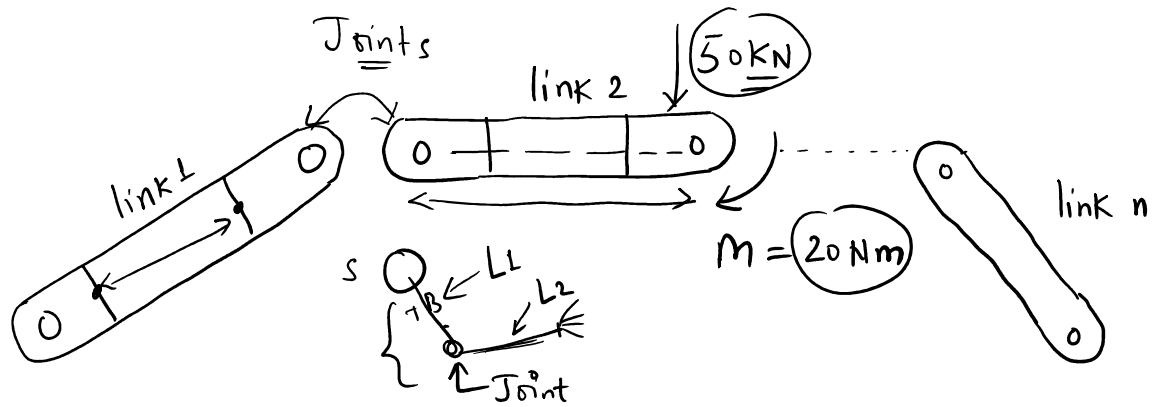
Robot — Links , Joints , Geometric representation,
Kinematic / Symbolic representation and
Kinematic diagram.

Topics Covered

- ① Links
- ② Joints or Kinematic Pairs
- ③ Kinematic Chain
- ④ Linkage
- ⑤ Mechanism
- ⑥ Machine
- ⑦ Geometric Representation of Robots.
- ⑧ Symbolic Representation.
- ⑨ Kinematic Representation.
- ⑩ Kinematic Diagrams.

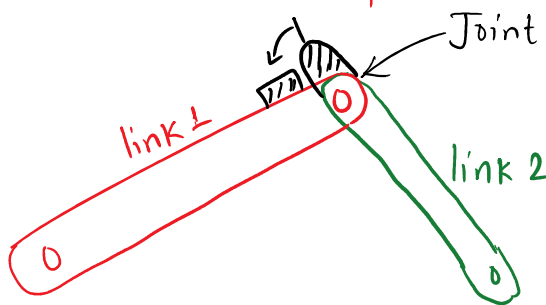
Robot Links

Link

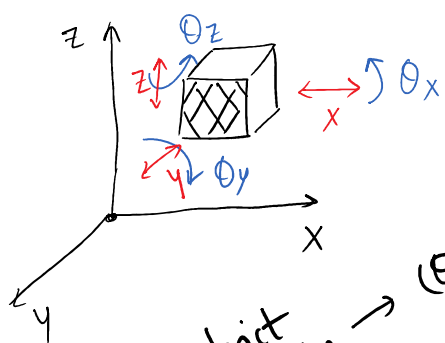
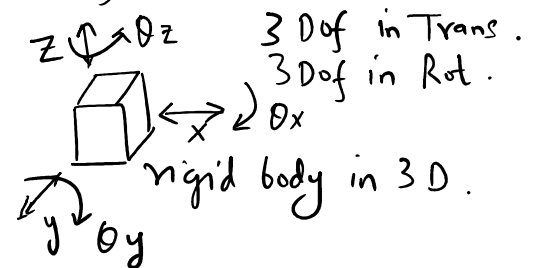


Links are the members (or elements) that forms the Robot. Connected in different configurations.

Considered to be rigid / resistant bodies.
 Links Joined together kinematically forms Joints which behave like kinematic pairs.

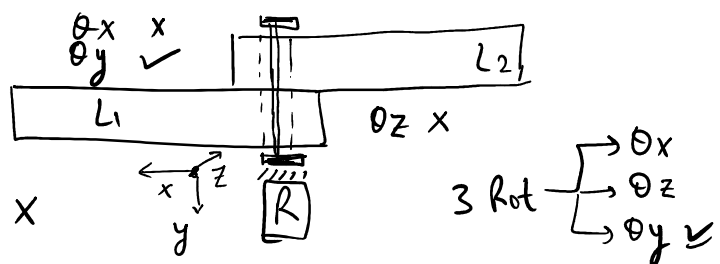
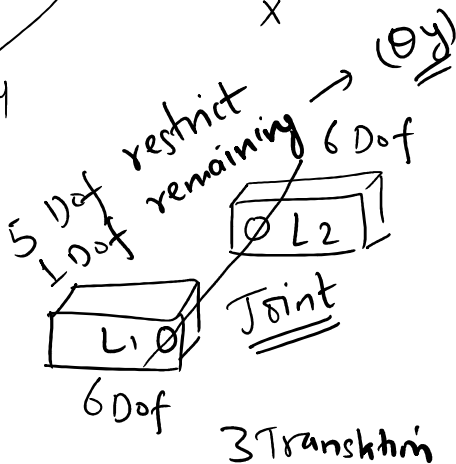


This joint puts physical constraint on the relative motion between the links).

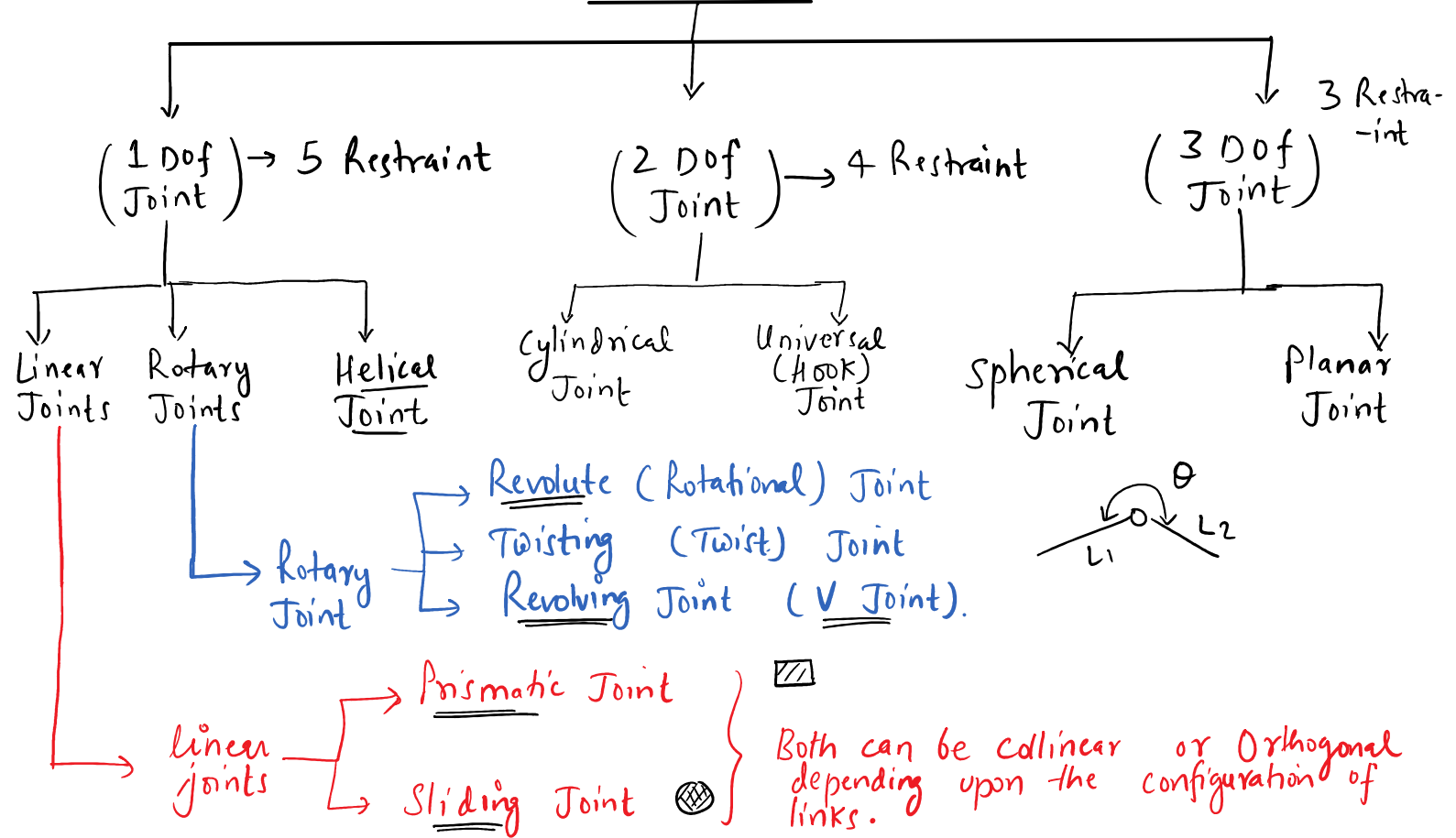


6 DOF's and (No Constraint)

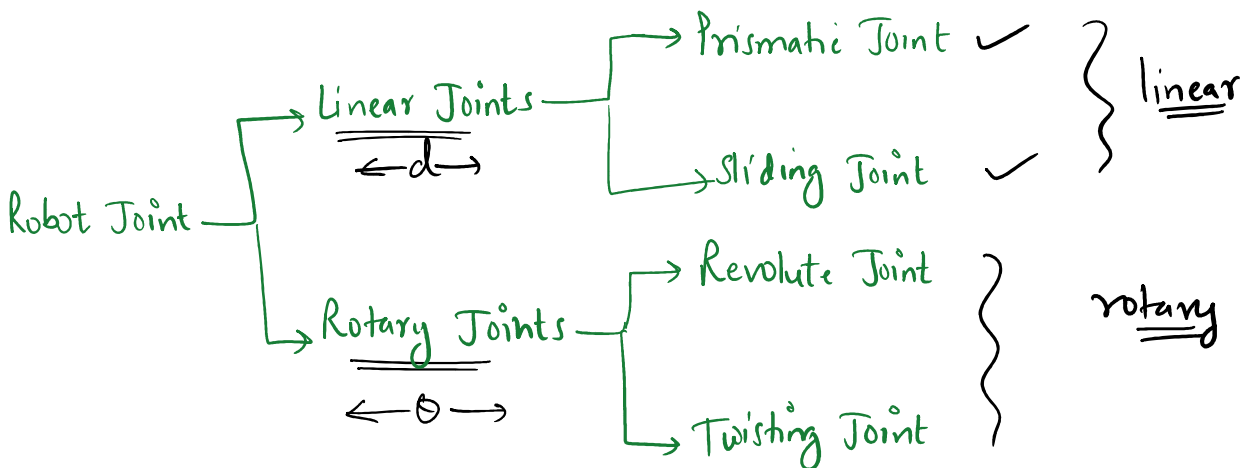
These joints helps put "constraints" on these movements and reduces the Dof to less than 6.



Robot Joints



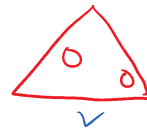
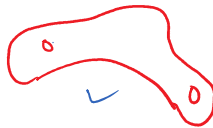
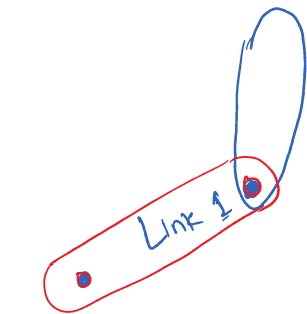
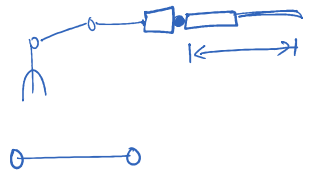
Generally, Robot's have two types of Joints only



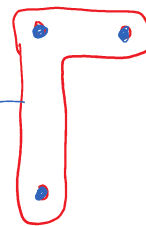
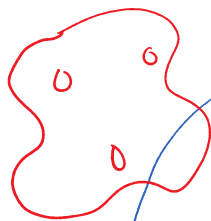
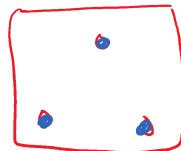
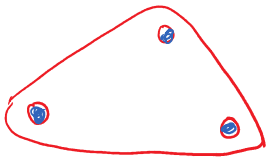
Kinematic chain

⇒ Link Types (kinematics)

- Singular link
- Binary link
- Ternary link
- Quaternary link

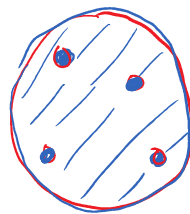
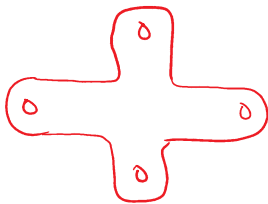
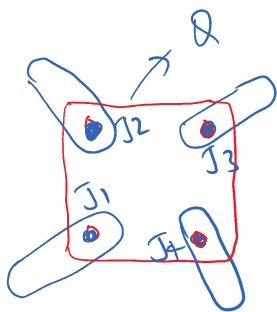
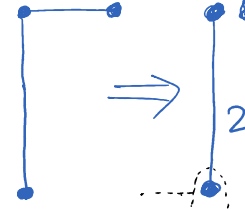
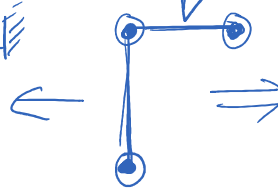
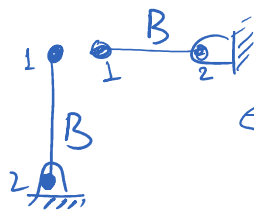


Binary links

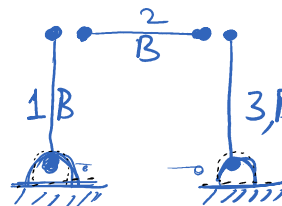
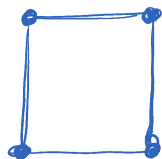
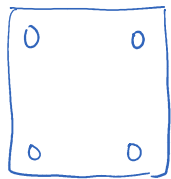


Ternary link == 2 Binary links

$$1T = 2B$$



Quaternary link = 3 Binary links

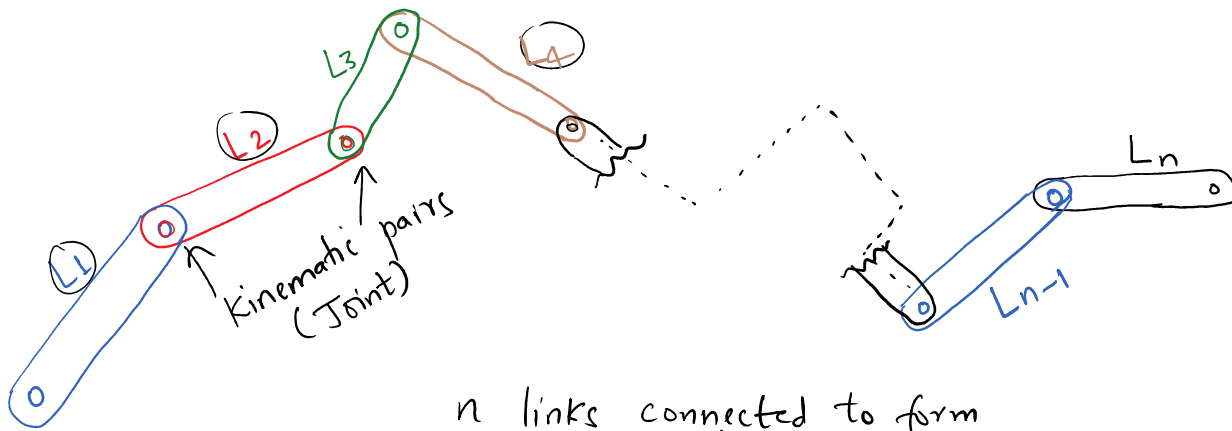


$$1Q = 3B$$



Kinematic chain : When these different links are connected in a series with the help of Joints to form Kinematic pair then the resulting arrangement of links and joints is known as Kinematic chain.

Kinematic Chain



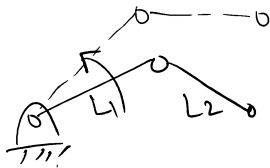
n links connected to form a kinematic chain.

Kinematic Chain: Series of links connected by kinematic pairs or joints. These k.c allows relative motion of one link w.r.t. other links and -this motion is definite.

Links Connected

(i) Kinematic Chains

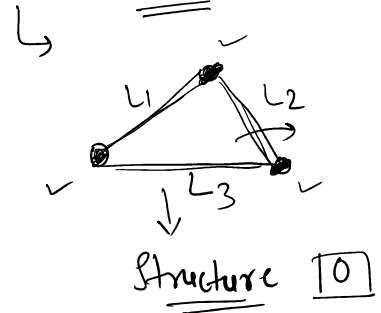
↳ Motion is defined and definite



(ii) Non Kinematic Chains

↳ Motion is indefinite

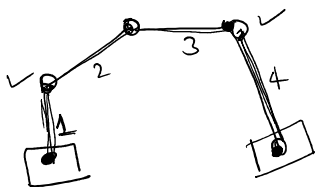
(iii) Redundant Chains



Kinematic Chains

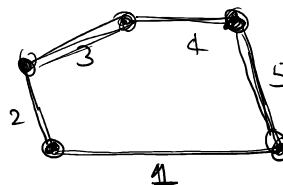
(i) Open Kinematic chain

every link is not connected to atleast two other links.



(ii) Closed Kinematic chain

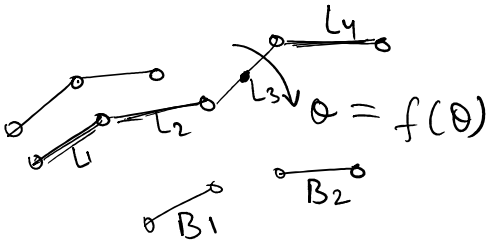
every link is connected to atleast two other links.



Kinematic chain

Kinematic Chain

(i) Simple K.C



(ii) Constrained K.C

A defined motion of one link produces unique relative motions of all other links.

[Hence, Dof = 1]

(iii) Redundant K.C

No relative motion is allowed.

Dof = 0

Structure = 0 X

Linkage : fix one link of a kinematic chain to the ground to get a linkage.

Definite Input Motion

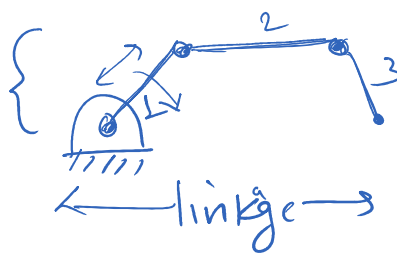


Definite motion of all other links

Mechanism.

Mechanism :

Definite θ, d



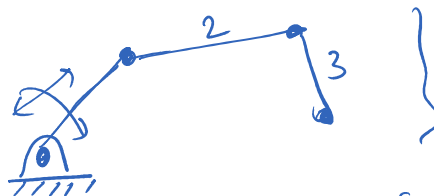
Output

Definite Motion of L2 & L3.

(Motion)

Machine :

Input
definite θ, d
etc



output, definite. of all other remaining links

power
input.

losses
Mechanism

output
80%.

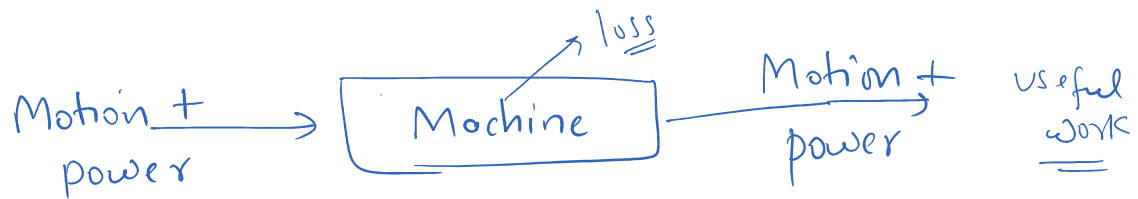
useful work

Machine

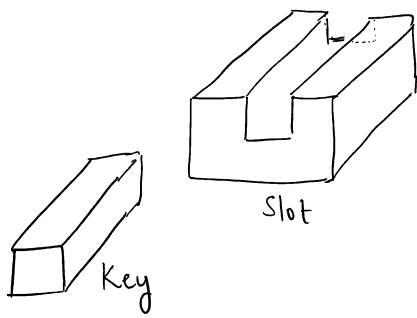
Mechanism & Machine

Input

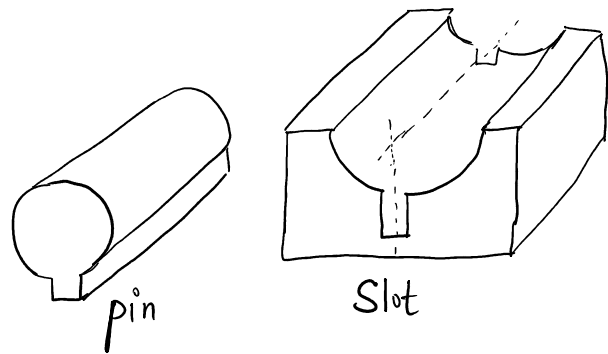
Output



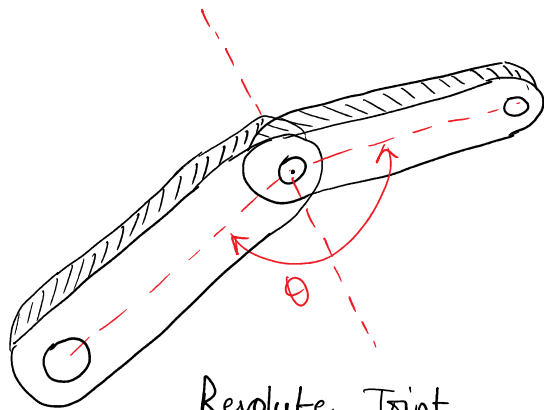
Geometric Representation



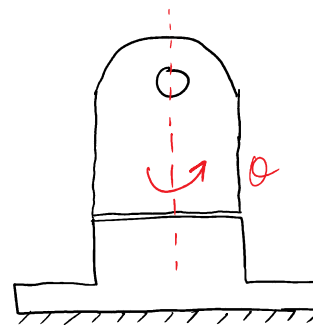
prismatic Joint



Sliding Joint



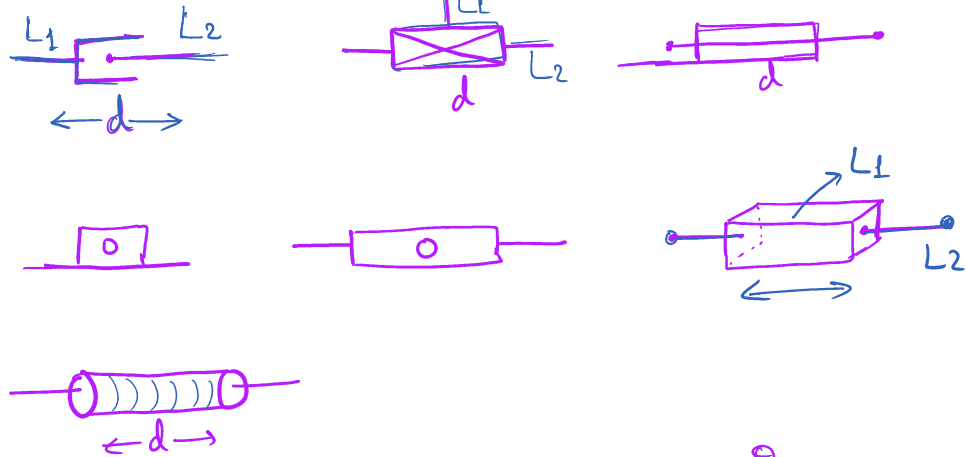
Revolute Joint



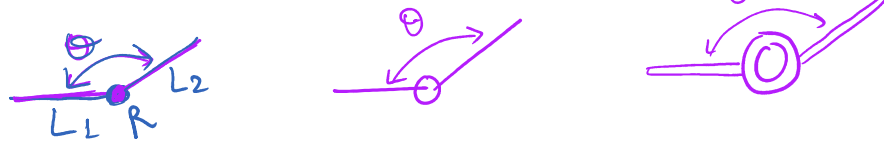
Twist Joint.

Joint - Symbolic Representation.

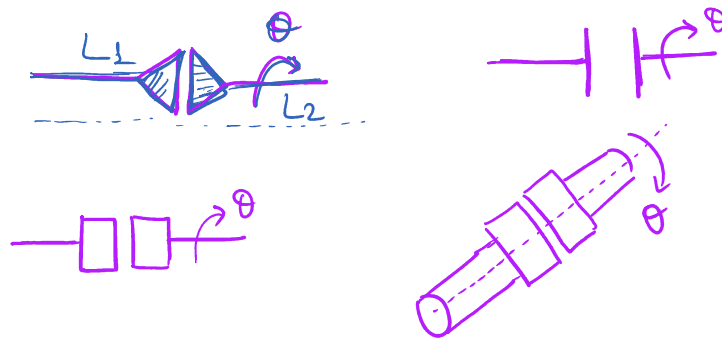
① Prismatic Joint, P :
and
Sliding Joint, S :



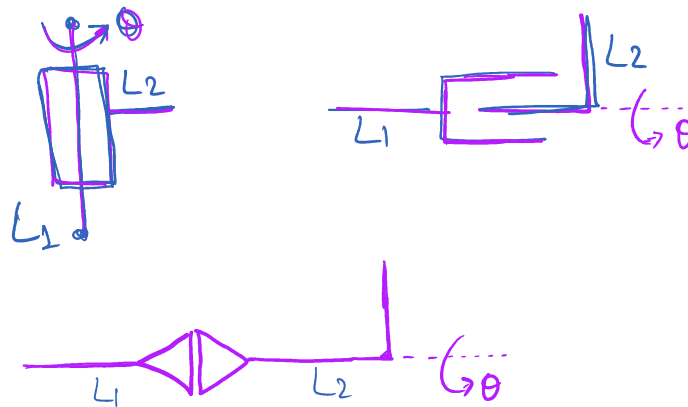
② Revolute Joint :
(R-Joint)



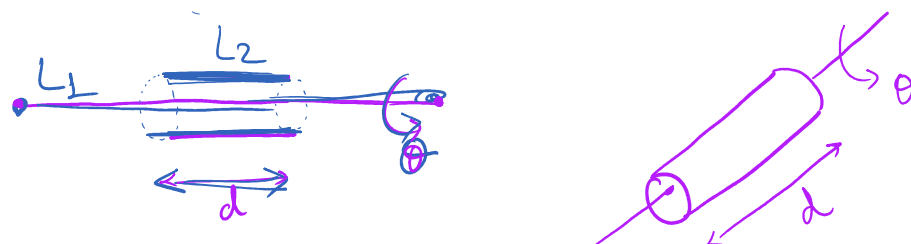
③ Twisting Joint :
(T-Joint)



④ Revolving Joint :
(V-Joint)

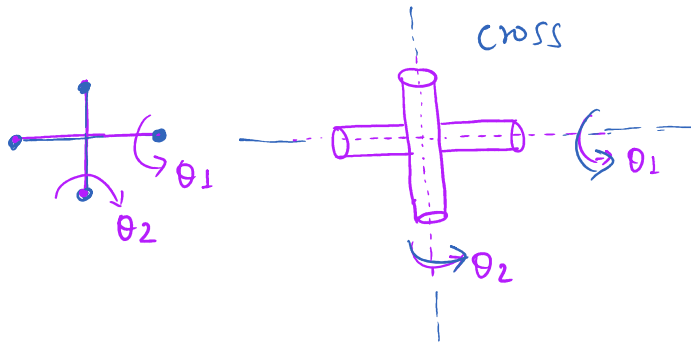


⑤ Cylindrical Joint :
(C-Joint)

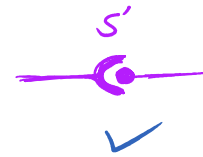
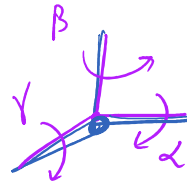
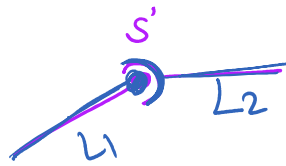


Joint - Symbolic Representation

⑥ Universal or Hook Joint :
(U-Joint)

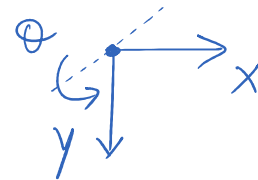
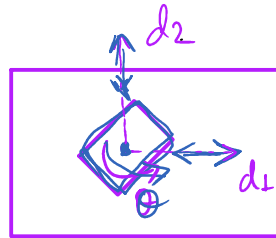


⑦ Spherical Joint :
(S'-Joint)
(S-prime)

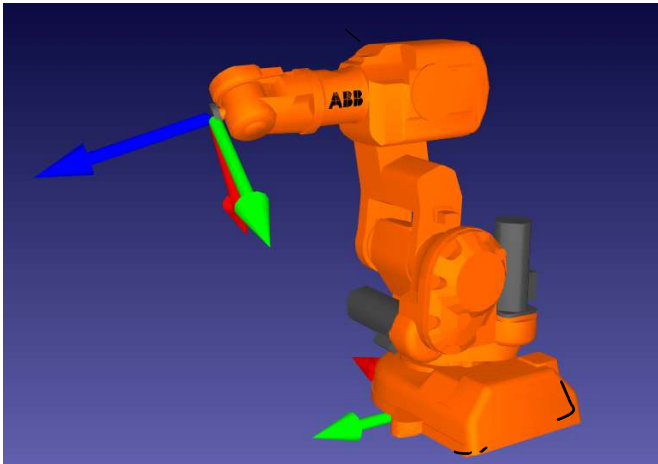
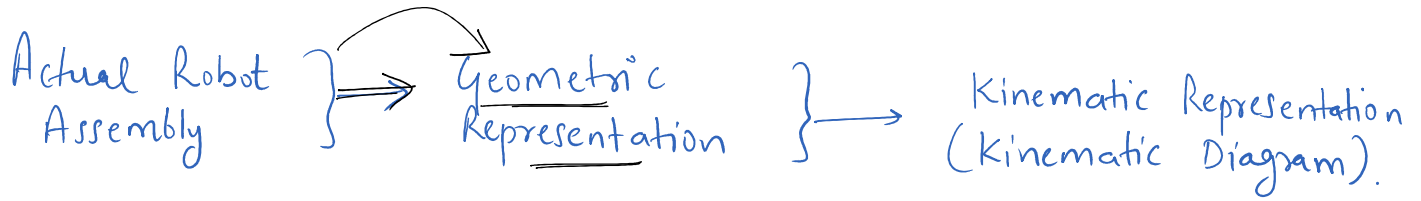


S → Sliding
S' → Spherical

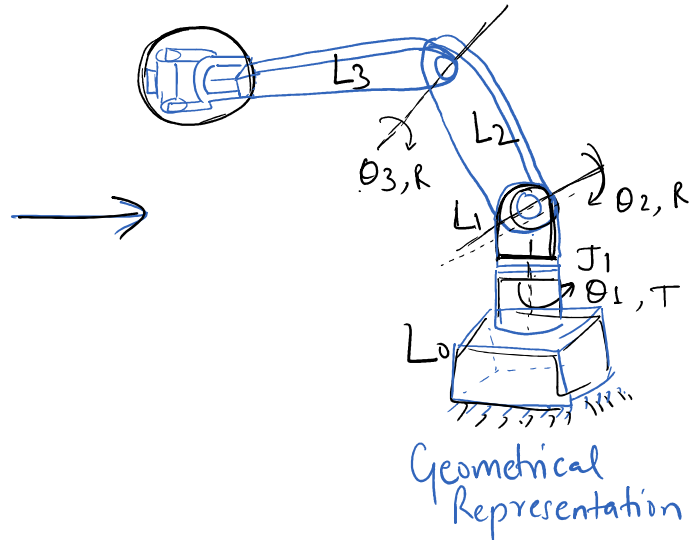
⑧ Planar Joint :



Kinematic Diagrams

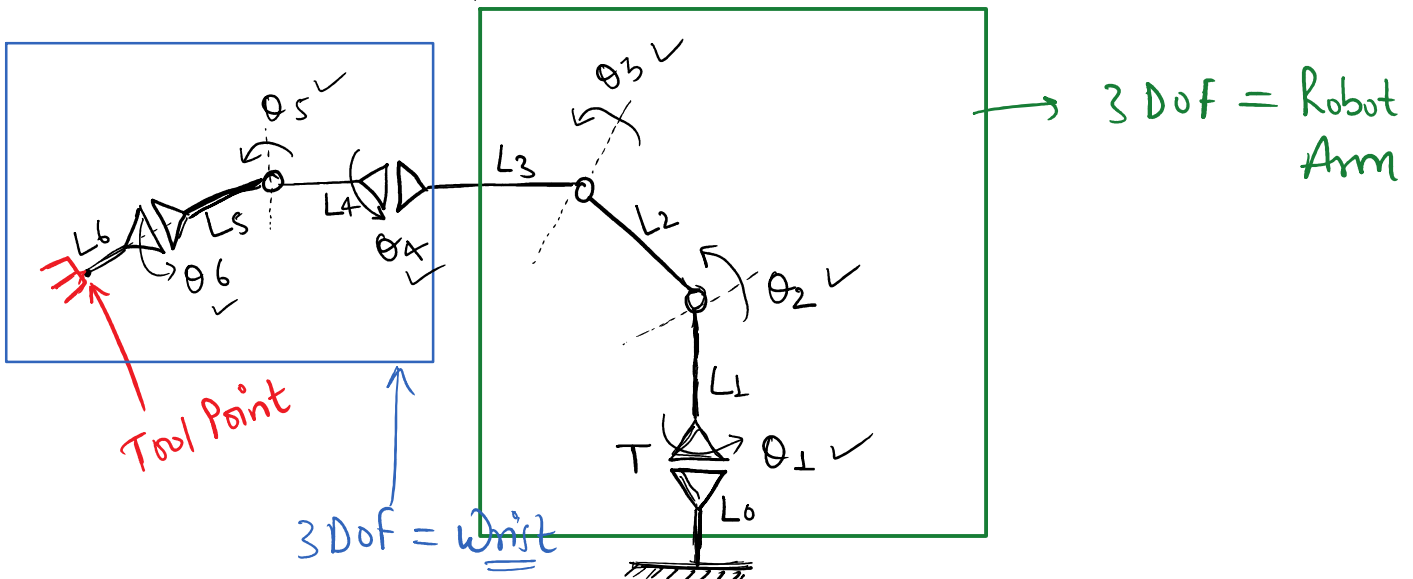


Actual Robot



Geometrical Representation

Kinematic Representation.



Robot Configurations

1 Cartesian Coordinate Configuration

2 Cylindrical Configuration

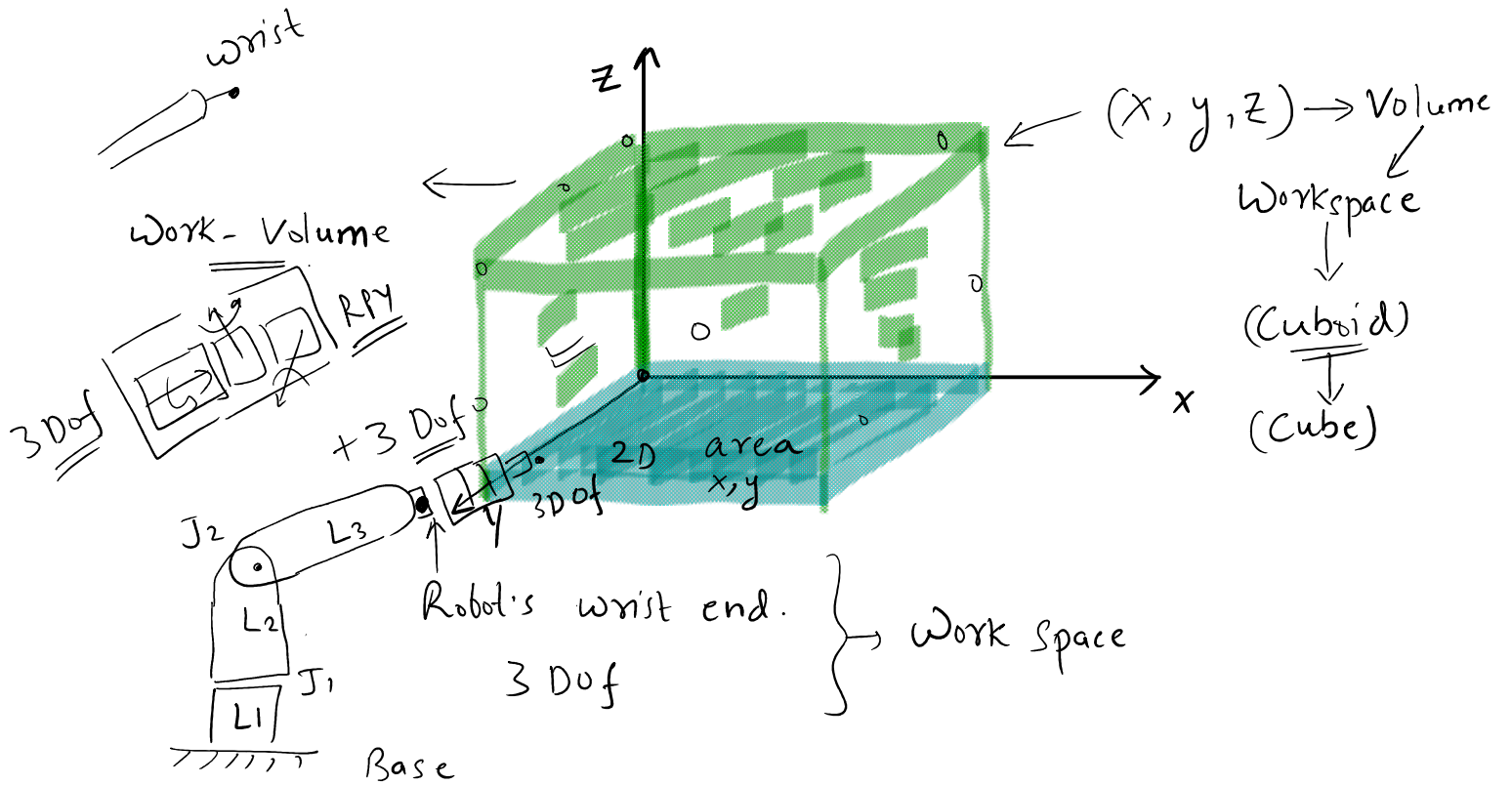
3 Spherical / Polar Coordinate Configuration

4 Jointed Arm / Articulated / Revolute Configuration

5 Selective Compliance Assembly Robot Arm / SCARA

6 and many more ...

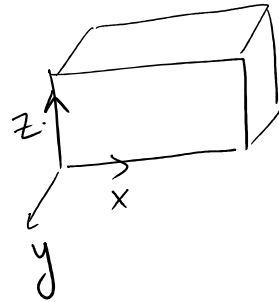
Cartesian / Rectangular Configuration.



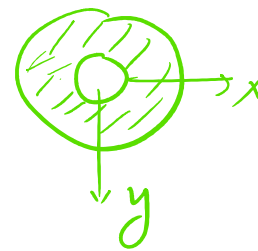
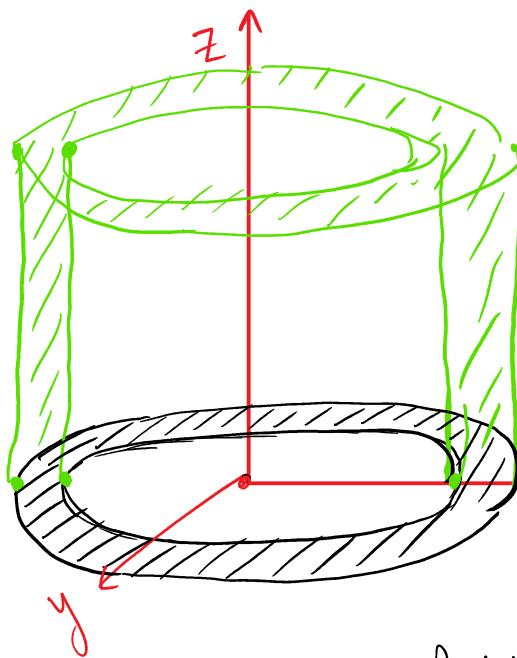
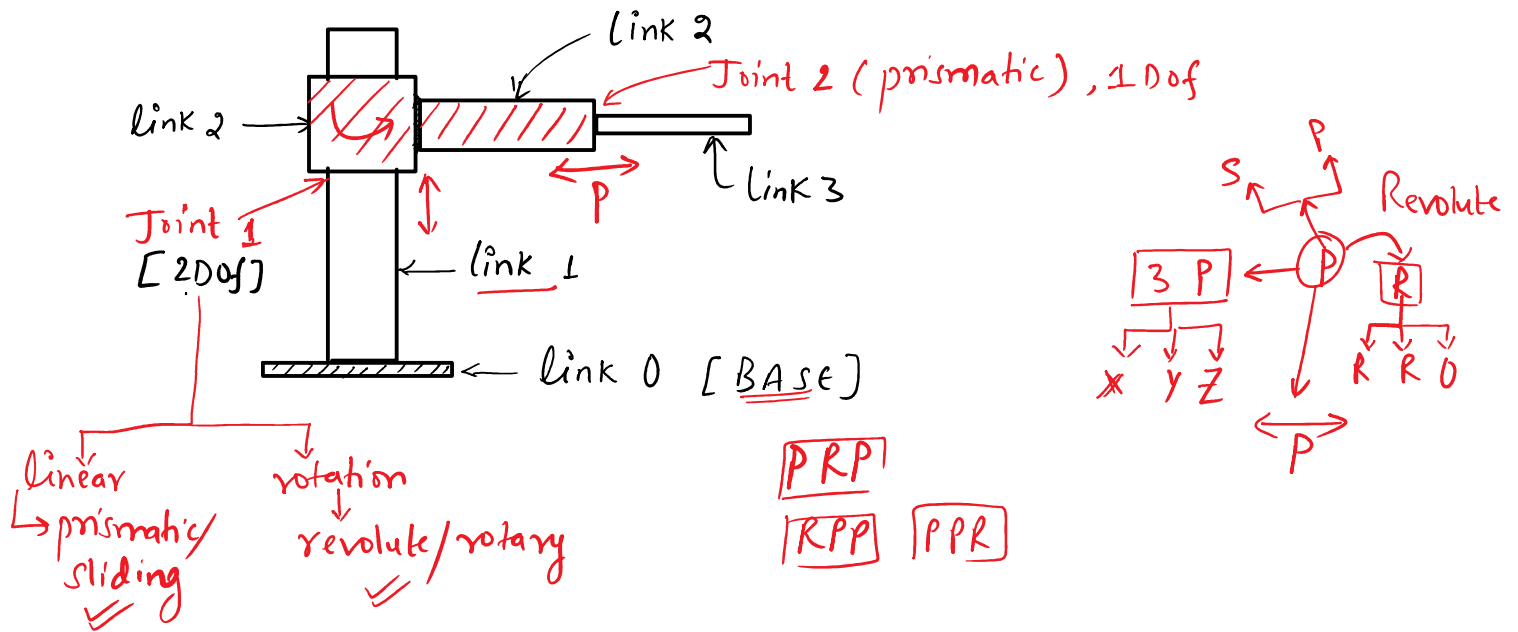
$(x, y, z) \rightarrow$ Volume
 Workspace
 ↓
 (Cuboid)
 ↓
 (Cube)

Volume } Work Volume $\geq$ Work Space

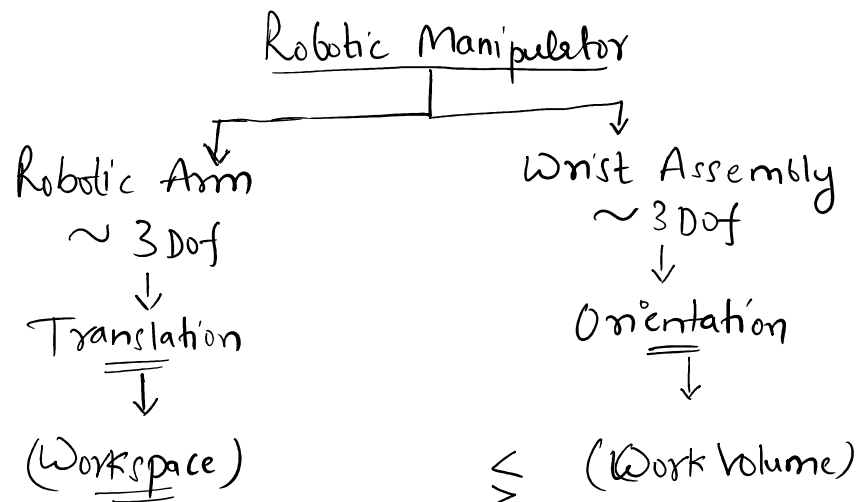
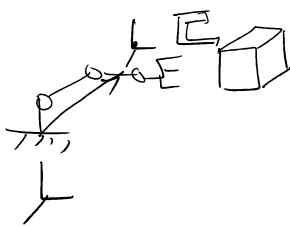
Workspace $\rightarrow$ { Cube
Cuboidal



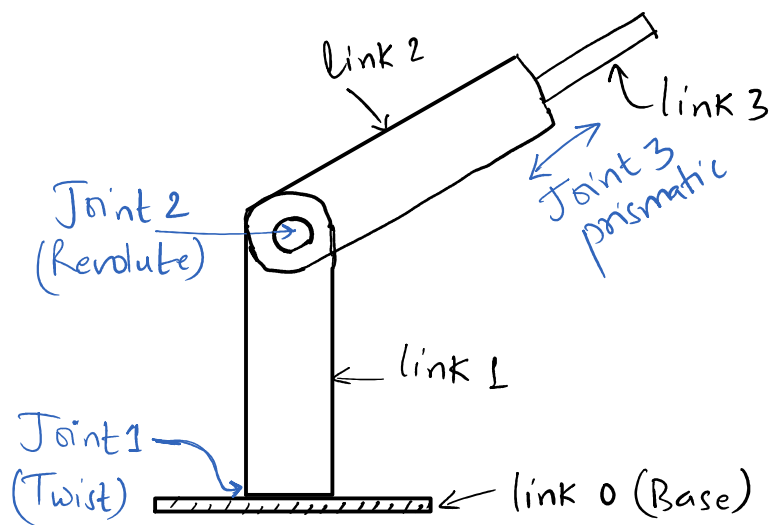
Cylindrical Configuration :



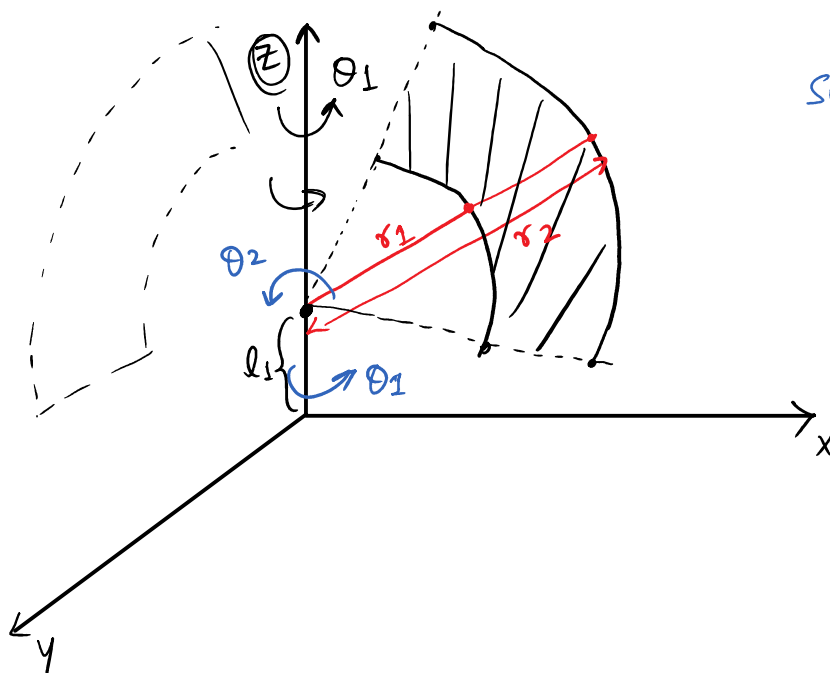
Workspace
↓
{ Concentric cylindrical volume }



spherical [Polar Coordinate] configuration:

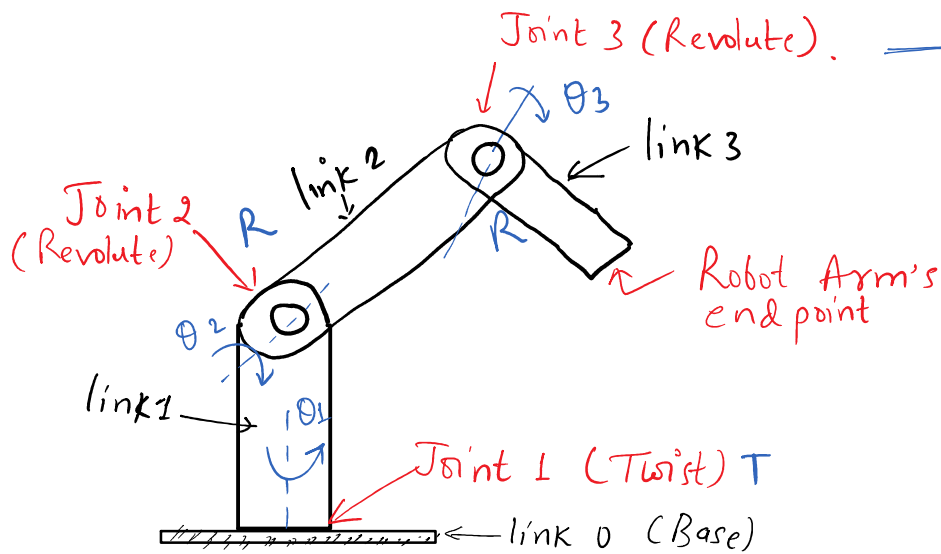


(TRP) or (TRS)
or
(RRP) or (RRS)
etc



Spherical coord' sys.
↓ ↓ ↓
 r θ_1 θ_2
↑ ↑ ↑
 $\{r, \theta, \phi\}$
↓

Jointed Arm / Articulated / Revolute Configuration.



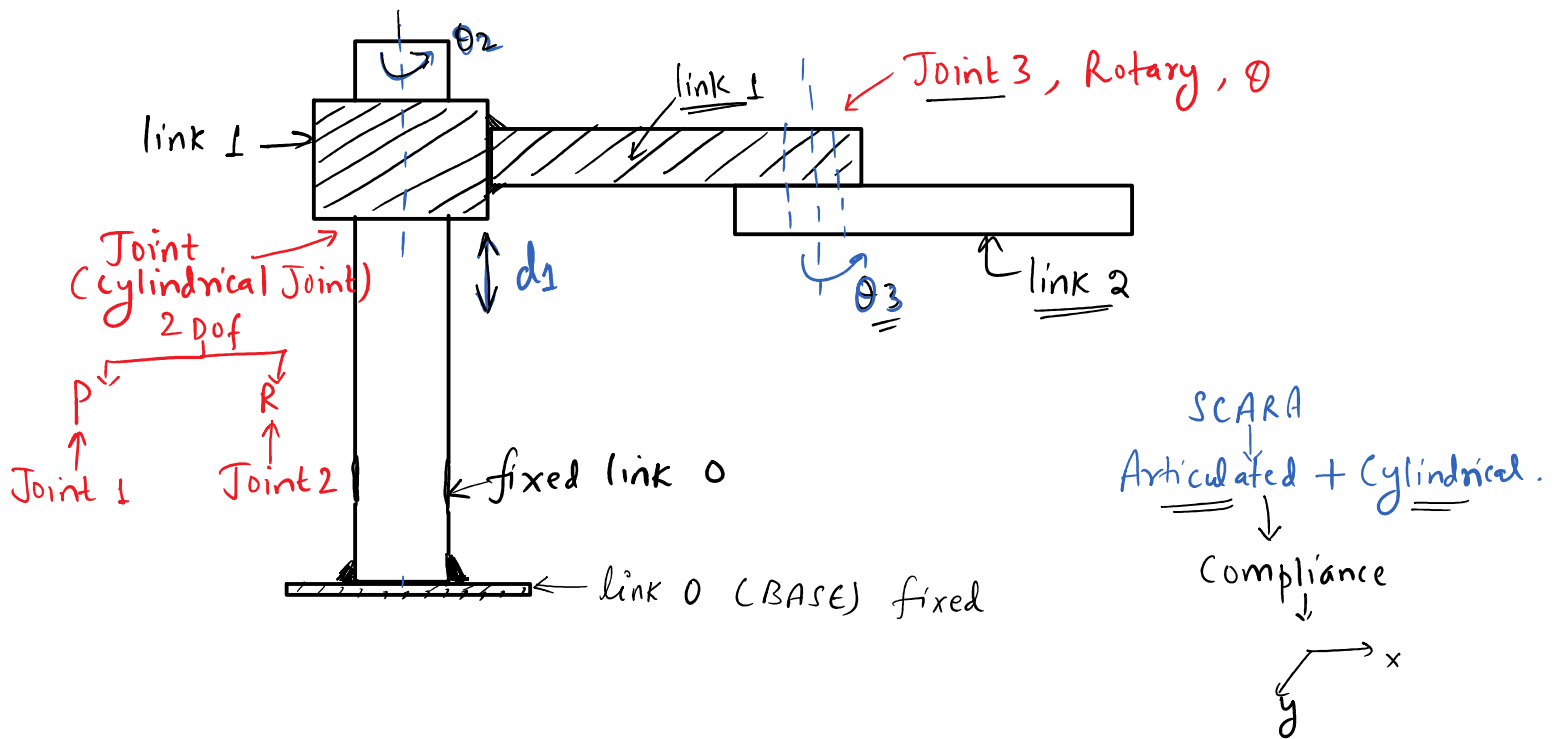
→ (TRR) or (RRR)

↓
{ Anthropomorphic Manipulator }

↓
[Human Arm]

workspace → Part of spherical volume.

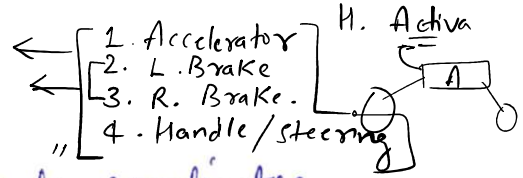
Selective Compliance Assembly Robot Arm (SCARA) Configuration



Degree of freedom (Connectivity)

Degree of freedom :

Dof $\begin{matrix} \textcircled{4} \\ \text{---} \\ \boxed{3} \end{matrix}$



The minimum number of independent coordinates (variables) required to fully define the configuration.

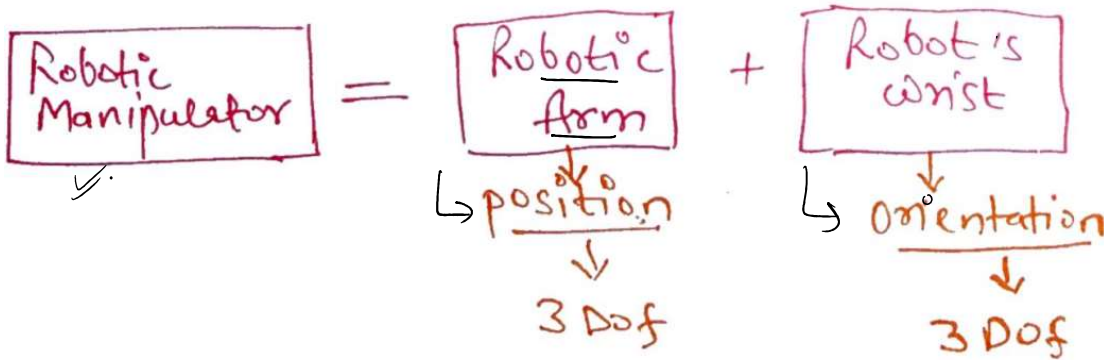
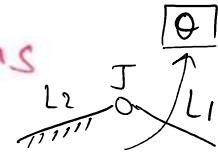
$L+R$

$\begin{matrix} L+R \\ R+L \end{matrix}$

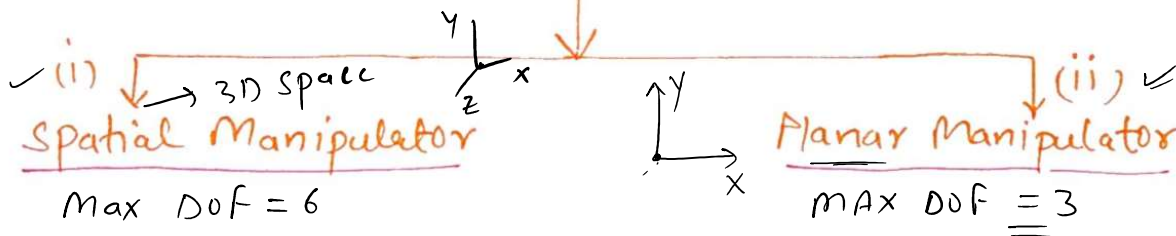
$\begin{matrix} \text{Combi} \\ \text{brake} \end{matrix}$

The number of "independent" movements that an object can perform in 3D space.

No. of independent relative motions a kinematic pair can have is known as its Degree of freedom.



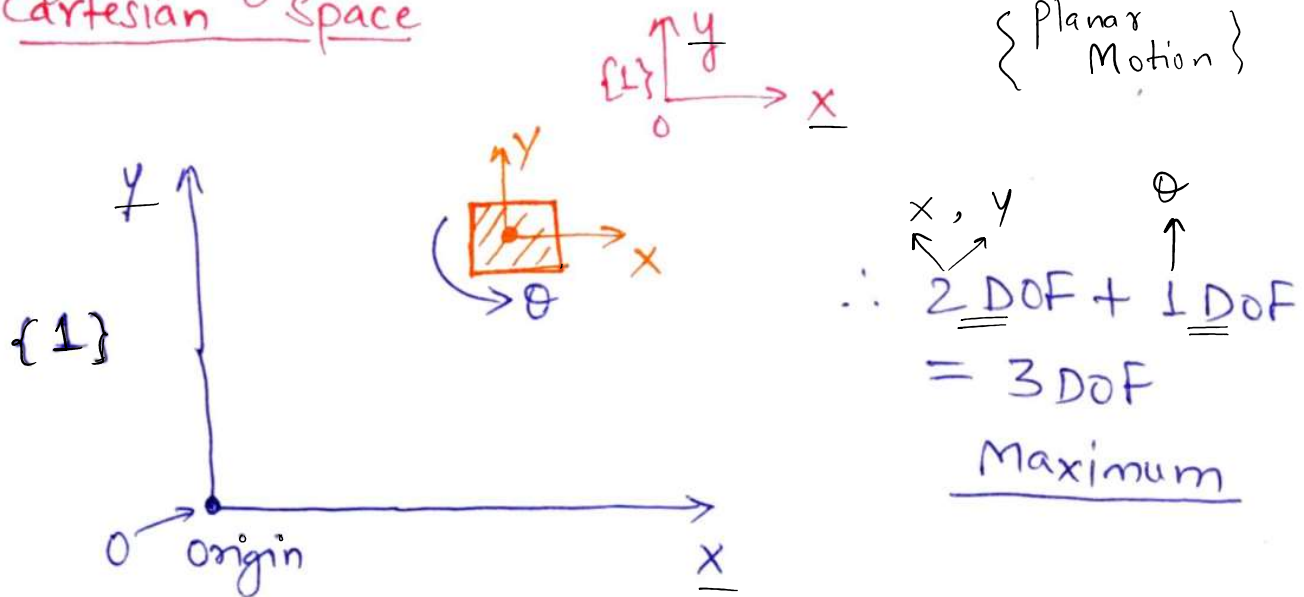
Manipulator Types Based on the number of "Dof"



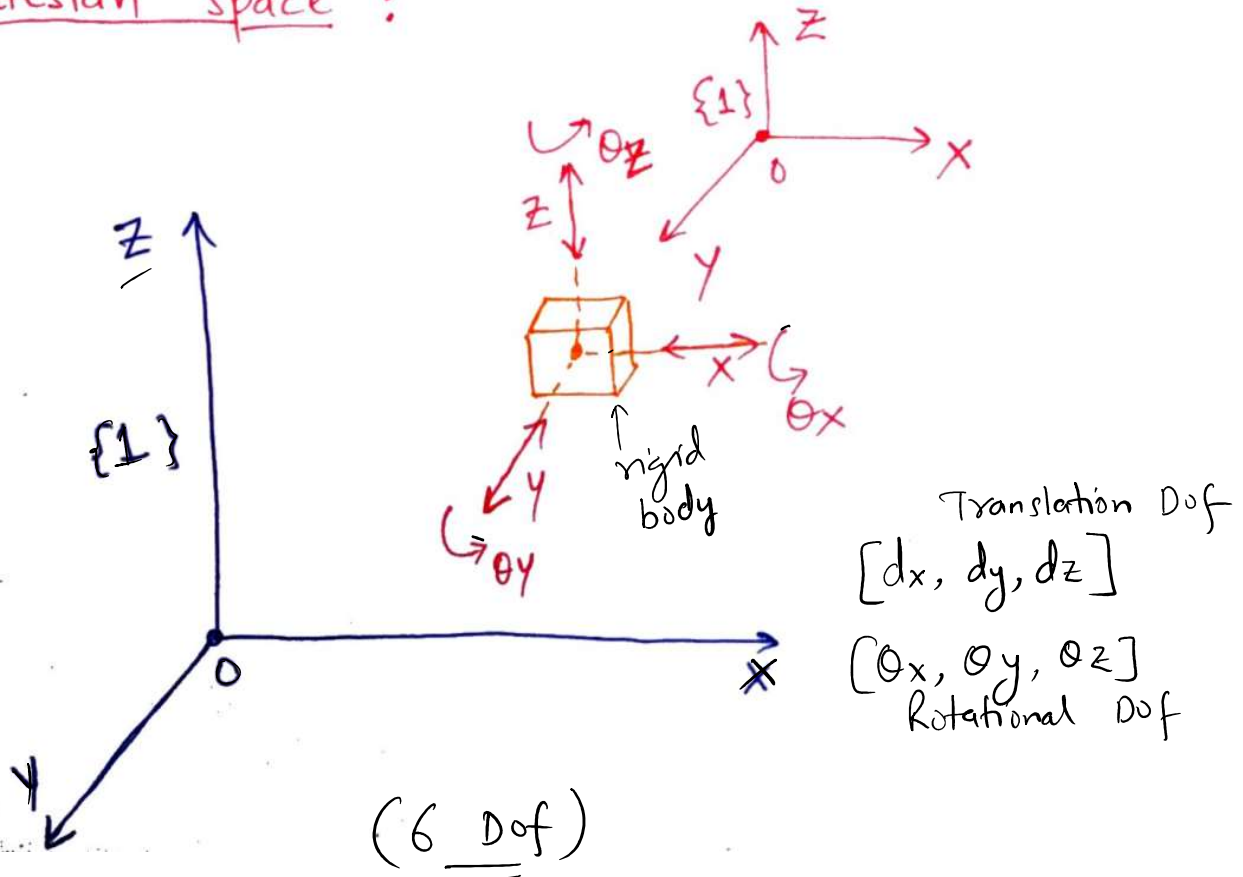
Redundant Manipulator $\rightarrow$ Dof > 6 (Dexterity) $\uparrow$

Rigid Body Motion.

Rigid body motion in 2 Dimensional Cartesian space

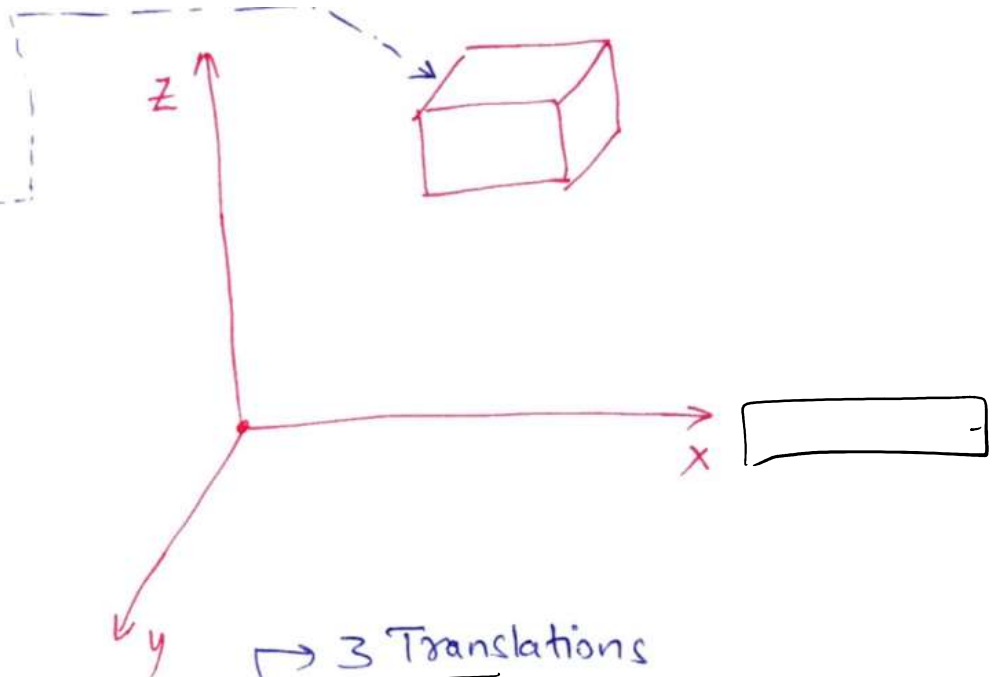


Rigid body motion in 3 Dimensional Cartesian space :



Continued.

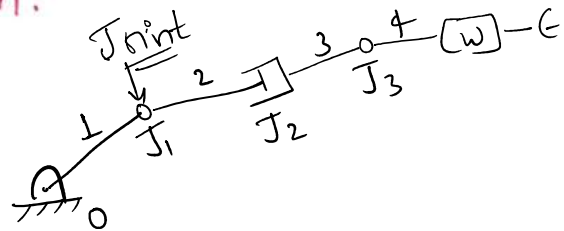
"unconstrained
rigid body in
three dimensions"



Its motion consists of — $\left\{ \begin{array}{l} \rightarrow 3 \text{ Translations} \\ \rightarrow 3 \text{ Rotations} \end{array} \right.$

$\therefore$ A rigid body (in 3 Dimensions) possesses
Six degree of freedom.

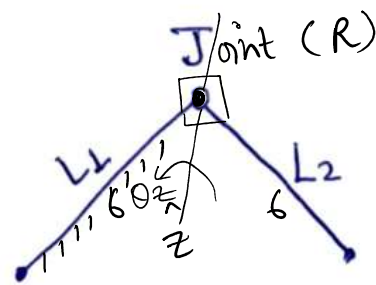
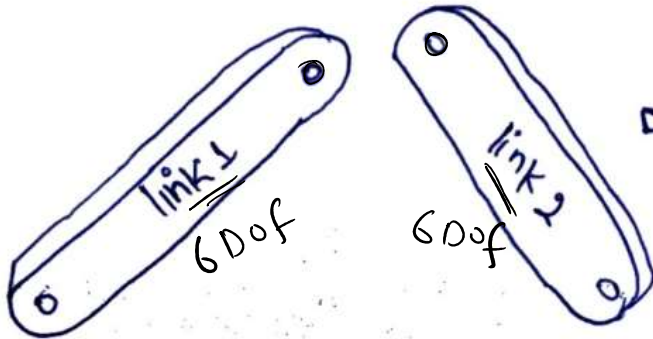
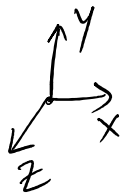
Kinematic Pairs (Joints)



When two rigid body connects together and forms a Joint (Kinematic pair).
 So Joint will always bring constraint with it.

Restraint

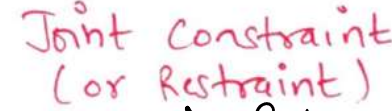
Joint



$$\begin{bmatrix} dx \rightarrow x \\ dy \rightarrow x \\ dz \rightarrow x \end{bmatrix}$$

$$\begin{bmatrix} \theta_x \rightarrow x \\ \theta_y \rightarrow x \\ \theta_z \checkmark \end{bmatrix}$$

6Dof $\begin{cases} \rightarrow 5 \text{ Constraint} \\ \rightarrow 1 \text{ Dof is left} \end{cases}$





2 Joint becomes Solid

$\perp C$, [Dof = 5]

2 C, [Dof = 4]

3 c, | (Dof = 3)

4C, (Dof = 2)

$\bar{S}C$ (Dof = 1)

0 [Not possible]

6 [Not possible]

↳ 6 constraint

$$= x:$$

$100f = 1$

1

$\frac{1}{p}$

2

$\frac{6}{2} \rightarrow 3$

u

A diagram showing a point source S' emitting waves through a slit of width 3λ . The slit is represented by a horizontal line with a central opening. The source S' is located above the slit, and the width of the slit is labeled as 3λ .

P'



3

A diagram of a cylindrical capacitor. It consists of two concentric cylinders. The outer cylinder is labeled with length c . The inner cylinder is labeled with diameter d . The diagram is labeled with a '2' in the bottom right corner.

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Expression for Degree of freedom {Mobility}

$$DOF = 6 - (\text{No of Constraint})$$

$$DOF = 3 - (\text{No of Constraint})$$

Expression for Degree of freedom

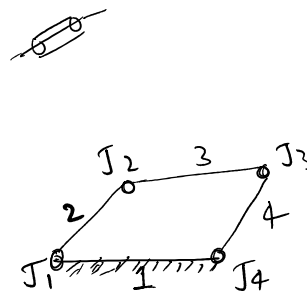
let

Total Number of links in a Mechanism $\} = L = 4$

DOF = Degree of freedom of this Mechanism.

$$(L-1)$$

$$4-1 = \textcircled{3}$$



P_1 = Number of ^{Joint} Kinematic pairs having one DOF = 4

P_2 = Number of Kinematic pairs having two DOFs = 0

P_3 = " " " " three DOFs = 0

P_4 = " " " " 4 DOFs = 0

P_5 = " " " " 5 DOFs = 0

since, in a Mechanism, one link is fixed to the ground.

$\therefore$ No. of "movable" links = $(L-1) = (4-1) = 3$

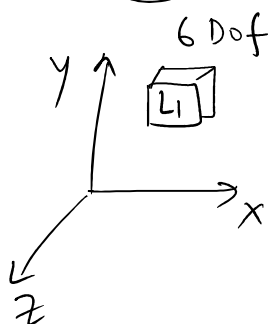
$$\boxed{L} \quad \textcircled{1}$$

$$(L-1)$$

$\rightarrow$ Movable

If one link has 6 DOF (spatial)

$\therefore$ $\textcircled{(L-1)}$ movable links have $6(L-1)$ DOFs



1 Link $\rightarrow$ 6 DOF

$(L-1)$ links $\rightarrow 6 \times (L-1)$

In planar Mechanism.

1 Link $\rightarrow$ 3 DOF

$(L-1)$ links $\rightarrow 3 \times (L-1)$

$$\begin{array}{l} L \rightarrow 4 \\ L_m \rightarrow 4-1=3 \\ 3 \times (L-1) \\ = 3 \times (3) \\ = 9 \end{array}$$

Continued

So, one Joint of 1 Dof puts 5 Restraints on the Mechanism

$\therefore P_1$ Joints will put $5 \cdot P_1$ restraints on the Mechanism.

$$5(P_1)$$

$$\boxed{5 \cdot P_1}$$

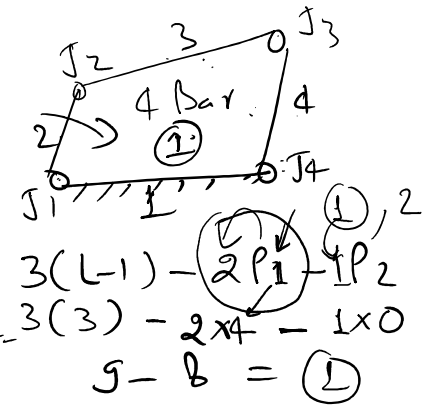
also, one Joint (Kinematic pair) of 2 Dof puts 4 Restraints on the Mechanism.

$\therefore P_2$ Joints will put $4 \cdot P_2$ restraints on the Mechanism

$$\boxed{4 \times P_2}$$

and so on. like

3 Dof, 4 Dof and 5 Dof joints will put 3, 2, 1 restraints respectively



$$Dof = 3(L-1) - 2P_1 - 4P_2$$

$$Dof = 3(3) - 2 \times 4 - 1 \times 0$$

$$5 - 8 = -3$$

$$2 \text{ Dof } \boxed{4 C}$$

$$3 \text{ Dof } , 3 C$$

$$4 \text{ Dof } , 2 C$$

$$5 \text{ Dof } , 1 C$$

$$Dof = 6 - (\text{No of Constraint})$$

$P_1 = \text{no of joints having 1 Dof}$ } 5 Constraint

Valid for Spatial mechanism. [Kutzbach's Criterion]

$$(Dof) = 6(L-1) - 5P_1 - 4P_2 - 3P_3 - 2P_4 - 1P_5$$

Redundancy / Passive Dof ?

$P_2 = \text{No of Joints having 2 Dof}$
 $4 = \text{no of Constraint}$

Continued

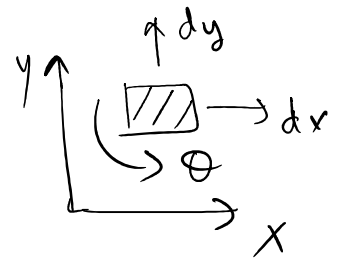
∴ Majority of Mechanisms in practice are of planar (or 2D) nature.

∴ In a plane, (Max Dof = 3) ✓

Total links (including ground) = L

Total movable links = $(L-1)$

Total Dof of movable links = $3(L-1)$



$$\boxed{L} = 4$$

$$(L-1) = \boxed{3}$$

$$6(L-1)$$

$$3(L-1)$$

$$\Rightarrow 3(L-1)$$

Continued

No of Joints of 1 Dof = $\underline{P_1}$ (let) = No. of Joints having 1 Dof

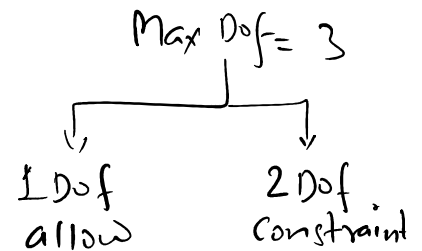
No of Joints of 2 Dof = $\underline{P_2}$ (let) = No. of Joints having 2 Dof

we know

1 Dof Joint puts = 2 restraints.]

$\therefore P_1$ Joints will put = $2P_1$ restraints

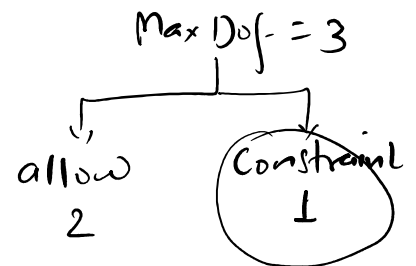
$$\Rightarrow 2 \times (P_1)$$



Similarly

2 Dof Joint puts = 1 restraint.]

$\therefore P_2$ Joints will put = $1P_2$ restraints



$$\boxed{\text{Dof} = 3(L-1) - 2P_1 - 1P_2}$$

← Grubler's Criterion

$$\text{Dof} = \boxed{6}(L-1) - 5.P_1 - 4.P_2 - 3.P_3 - 4.P_2 - 5.P_1$$

(planar)

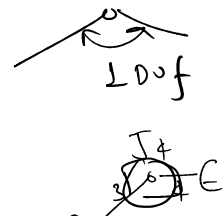
$$\boxed{\text{Dof} = 3(L-1) - 2.P_1 - 1.P_2}$$

← Grubler's Criterion

Other Method of Calculating Dof.

$$Dof = \left\{ \begin{array}{l} \rightarrow 6 - (\text{No of constraint}) \\ \rightarrow 3 - (\text{No of constraint}) \end{array} \right\}$$

let, a manipulator has $\left\{ \begin{array}{l} \textcircled{L} \text{ links (rigid)} \\ \textcircled{J} \text{ Joints} \end{array} \right\}$ (Mobile)



let for any joint i .

Connectivity of i^{th} Joint = $\boxed{C_i}$ $\rightarrow$ $\boxed{Dof}$

Restraint of i^{th} Joint = $\boxed{6 - C_i}$

$\left\{ \begin{array}{l} \rightarrow (6 - C_i) \\ \rightarrow (3 - C_i) \end{array} \right\}$ Constraints

$$L = \boxed{4}$$

$$J = \boxed{4}$$

$\therefore$ Total no of constraints of all the J Joints = $\sum_{i=1}^J (6 - C_i)$

$$\sum_{i=1}^J (6 - C_i)$$

$$\therefore \text{Dof or Mobility} = 6L - \sum_{i=1}^J (6 - C_i)$$

also $\text{Dof} = 3L - \sum_{i=1}^J (3 - C_i)$

Constraint of the i^{th} Joint

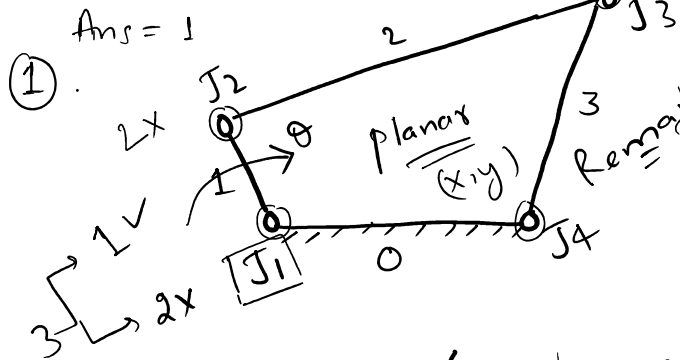
$$(Dof) = 6L - \sum_{i=1}^J (6 - C_i) \quad \checkmark$$

$\Rightarrow$ On expansion...

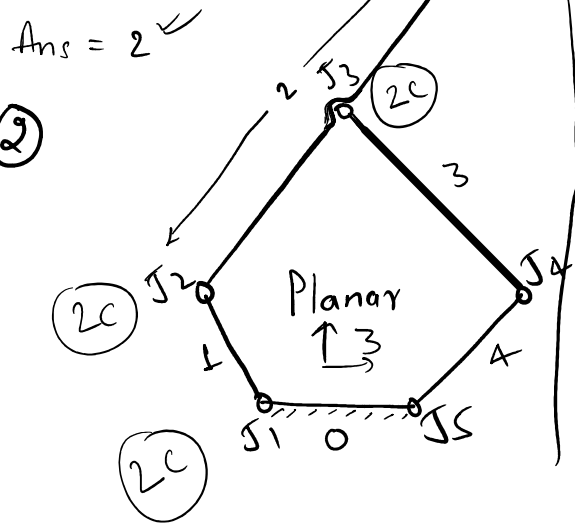
$$(Dof) = 6L - 5P_1 - 4P_2 - 3P_3 - 2P_4 - 1P_5$$

$\uparrow$ $L = \text{Total no of movable links}$

Solved examples on Dof



$$\text{Dof} = \left[\begin{array}{l} \text{Total Dof of} \\ \text{all the} \\ \text{movable} \\ \text{links} \end{array} \right] - \left[\begin{array}{l} \text{Total Constraints} \\ \text{on all the} \\ \text{Joints} \end{array} \right]$$

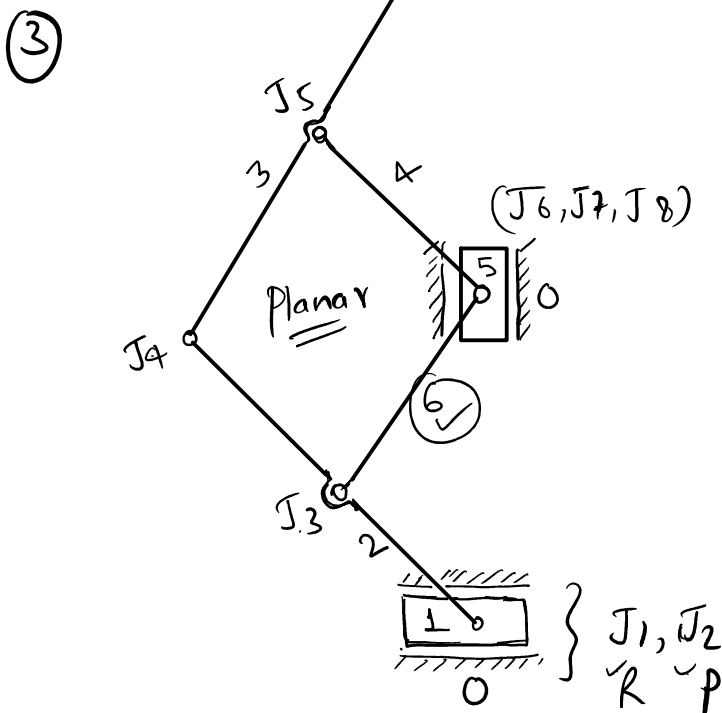


$$\begin{aligned} \text{Dof} &= 3L - 2P_1 - 1P_2 \\ &= 3 \times 4 - 2(5) - 1(0) \\ &= 12 - 10 \\ \text{Dof} &= 2 \end{aligned}$$

$$\text{Dof} = \left[\begin{array}{l} \text{Total Dof of all} \\ \text{the movable} \\ \text{links} \end{array} \right] - \left[\begin{array}{l} \text{Total constraints} \\ \text{on all the joints} \end{array} \right]$$

$$\begin{aligned} \text{Dof} &= 3(4) - \{2 + 2 + 2 + 2 + 2\} \\ \text{Dof} &= 12 - 10 = 2 \end{aligned}$$

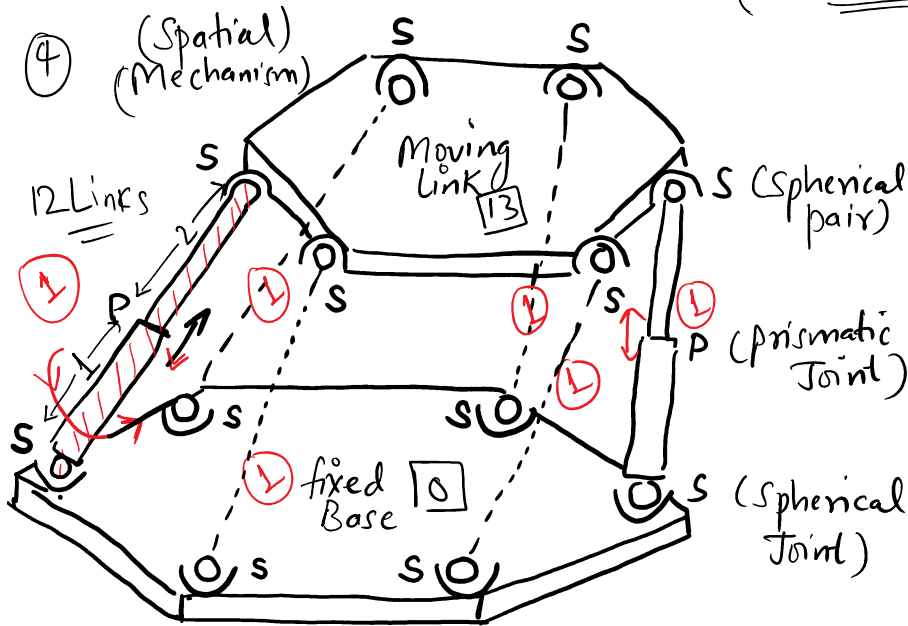
Ans = 2



$$\begin{aligned} \text{Dof} &= \text{Total Dof} - \text{Total Constraint} \\ &= 3(6) - \{2 \times 8\} \\ &= 18 - 16 \\ &= \underline{\underline{2 \text{ Dof}}} \end{aligned}$$

Solved examples on Dof

Ans = 6
(redundant Dof)



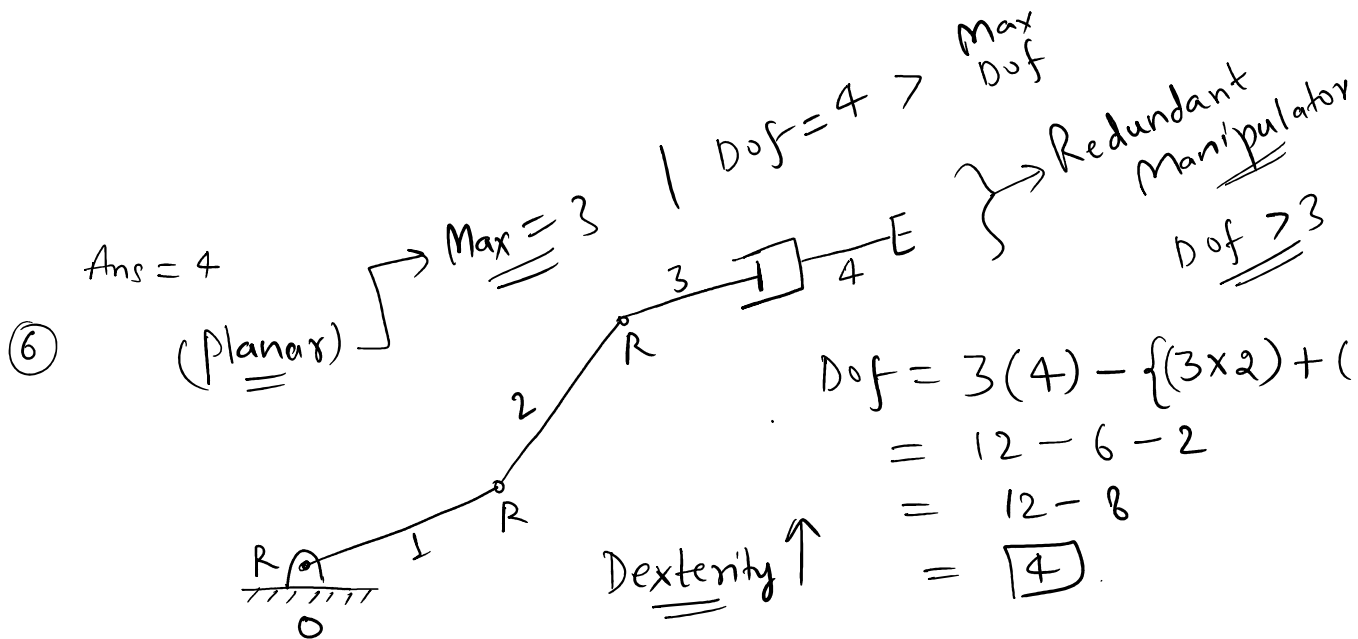
$$Dof = [6(13)] - \{(12 \times 3) + (6 \times 5)\} -$$

$$= 78 - 36 - 30$$

$$= 78 - 66$$

$$= 12 - 6$$

$$\Rightarrow Dof = \underline{\underline{6 \text{ Ans}}}$$

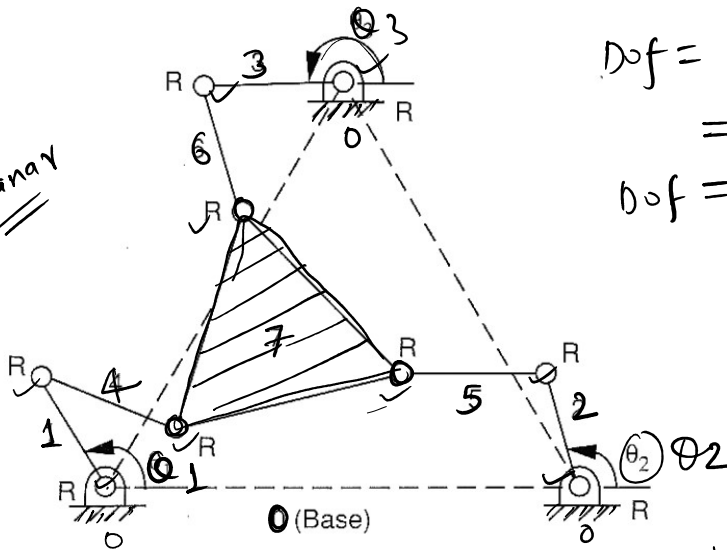


Solved examples on Dof

Ans = 3

⑦

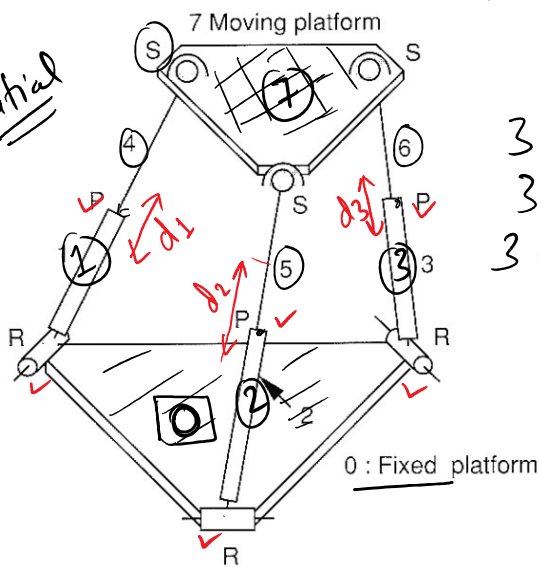
Planar



$$\begin{aligned} \text{Dof} &= 3(7) - \{9 \times 2\} \\ &= 21 - 18 \\ \text{Dof} &= 3 \text{ Dof} \end{aligned}$$

⑧

Spatial

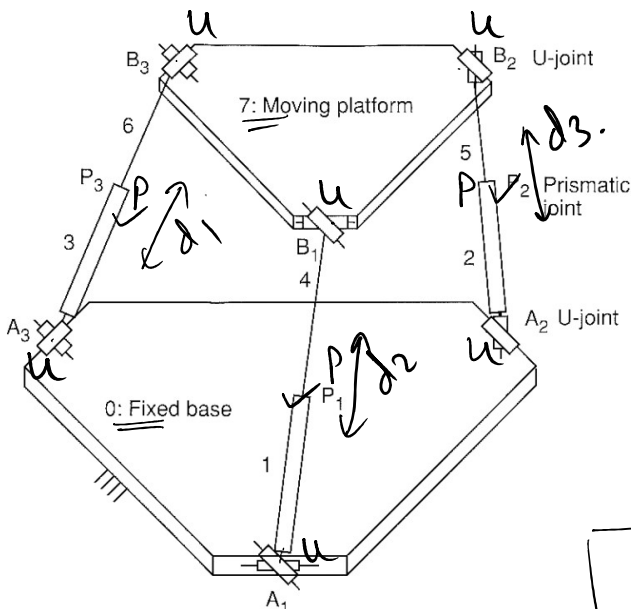


$$\begin{aligned} \text{Ans} &= 3 \quad | \quad \text{Dof} = 6(7) - \{9 + 15 + 15\} \\ &= 42 - 39 \end{aligned}$$

$$\begin{aligned} 3 \text{ Sph.} &\rightarrow 3 \times 3C = 9C \\ 3 \text{ prisms} &\rightarrow 3 \times 5C = 15C \\ 3 \text{ revol.} &\rightarrow 3 \times 5C = 15C \end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} = 3 \text{ Dof}$$

$$\text{Dof} = 6(N-1) - 5P_1 - 4P_2 - 3P_3 - 2P_4 - 1P_5$$

⑨



$$\text{Dof} = 6(N-1) - 5P_1 - 4P_2 - 3P_3 - 2P_4 - 1P_5$$

$$\begin{aligned} &= 6(8-1) - 5(3) - 4(6) \\ &\quad - 3(0) - 2(0) - 1(0) \end{aligned}$$

$$= 42 - 15 - 24 - 0$$

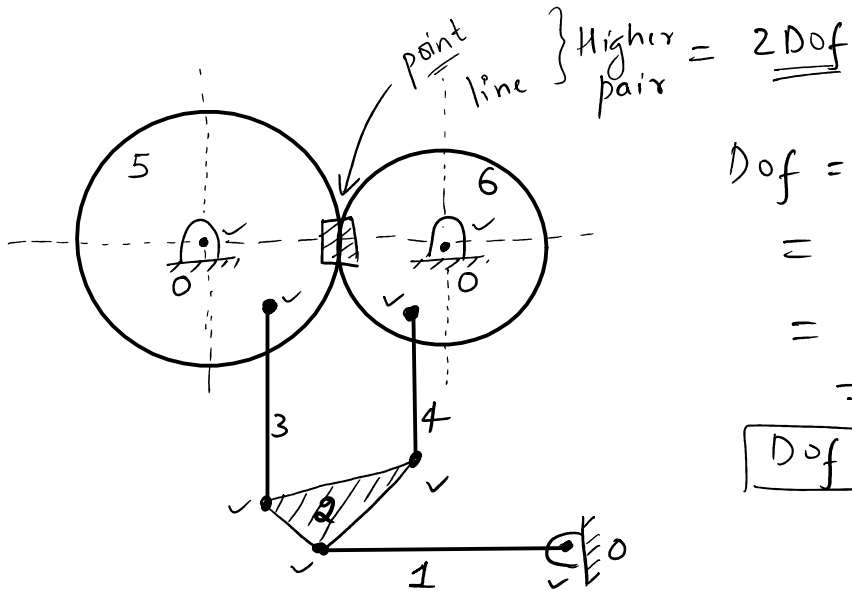
$$= 42 - 39$$

$$\boxed{\text{Dof} = 3}$$

Solved examples on Dof

Ans = 1

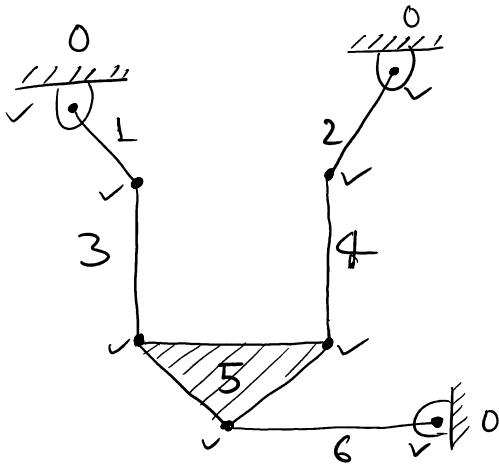
①



$$\begin{aligned} \text{Dof} &= 3(L) - 2P_1 - 1P_2 \\ &= 3(6) - 2 \times 8 - 1 \\ &= 18 - 16 - 1 \\ &= 18 - 17 \end{aligned}$$

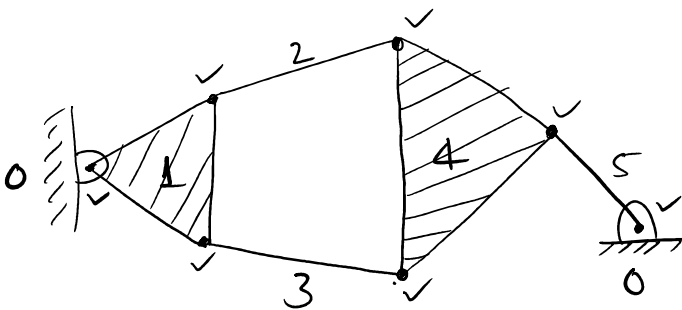
$$\boxed{\text{Dof} = 1}$$

②



$$\begin{aligned} \text{Dof} &= 3(L) - 2P_1 - P_2 \\ &= 3(6) - 2(8) - 0 \\ &= 18 - 16 \\ &= 2 \end{aligned}$$

3

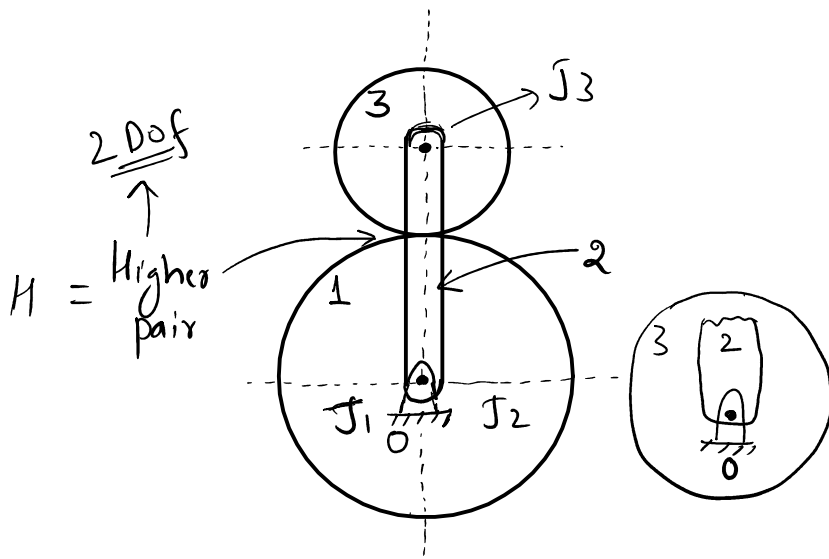


$$\begin{aligned} \text{Ans} &= 1 \\ \text{Dof} &= 3L - 2P_1 - P_2 \\ &= 3 \times 5 - 2 \times 7 - 0 \\ &= 15 - 14 \\ \text{Dof} &= 1 \end{aligned}$$

Solved examples on Dof

④

$$\text{Ans} = \underline{\underline{2 \text{ Dof}}}$$



$$\begin{aligned} \text{Dof} &= 3L - 2P_1 - 1P_2 \\ &= 3(3) - 2(3) - (1) \\ &= 9 - 6 - 1 \\ &= 9 - 7 \end{aligned}$$

$$\boxed{\text{Dof} = 2}$$

Redundant link } $\rightarrow$ If a link can be moved without causing any movement in the rest of the mechanism then such a link is known as a redundant link.

Redundant Dof } $\rightarrow$ The Dof associated with a redundant link is known as redundant Dof.

Let, In a Planar Mechanism,

Number of redundant links = $\boxed{L_r}$

Number of redundant kinematic pairs = $\boxed{P_r}$

Number of redundant Dof = $\boxed{F_r}$

Then,

$$(\text{Dof}) = 3(L_T - L - L_r) - 2(P_1 - P_r) - 1 \cdot P_2 - F_r$$

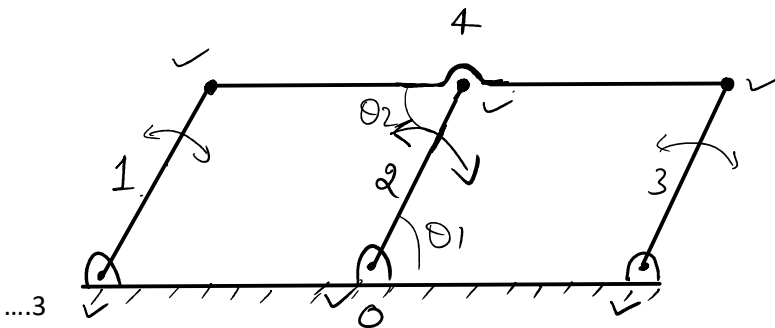
$$= 3(L_T - 1 - L_r) - 2(P_1 - P_r) - P_2 - F_r$$

Example on next page...

$$(Dof) = 3(L_T - L - L_r) - 2(P_1 - P_r) - 1 \cdot P_2 - F_r$$

$$\underline{\underline{Ans = 1}}$$

$$\begin{aligned} Dof &= 3(L) - 2P_1 - P_2 - F_r \\ &= 3(4) - 2(6) - 0 - 0 \\ &= 12 - 12 \\ &= \underline{\underline{0}} \end{aligned}$$

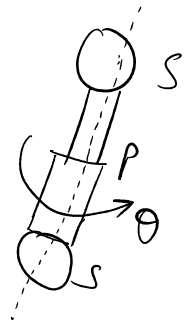


$$(Dof) = 3(L_T - L - L_r) - 2(P_1 - P_r) - 1 \cdot P_2 - F_r$$

$$\begin{aligned} Dof &= 3(5 - 1 - 1) - 2(6 - 2) - 0 - 0 \\ &= 3(3) - 2(4) \\ &= 9 - 8 \end{aligned}$$

$$\boxed{Dof = 1} \quad \underline{\underline{Ans}}$$

Red. link }
Red. Joint }
Red. Dof }



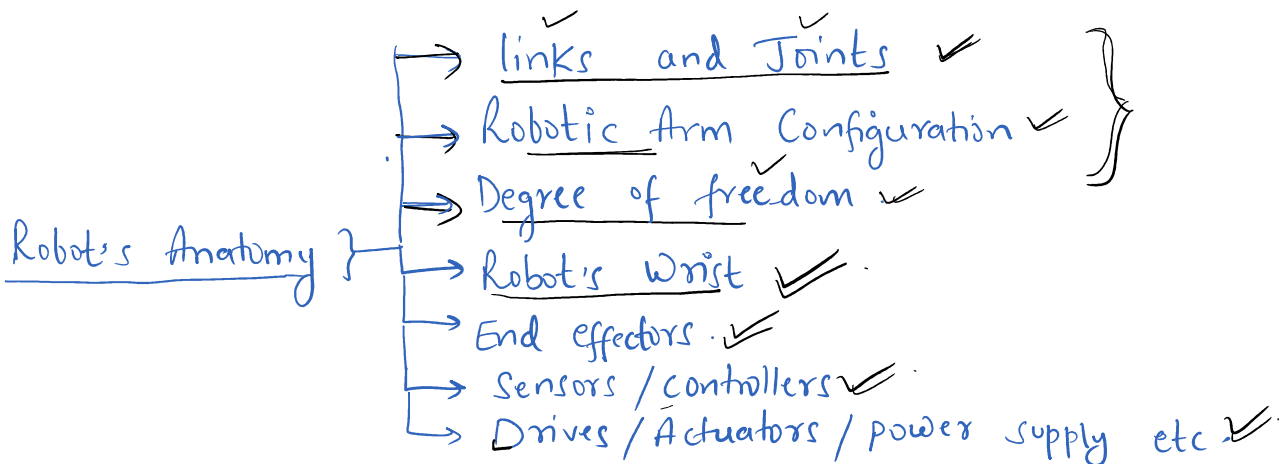
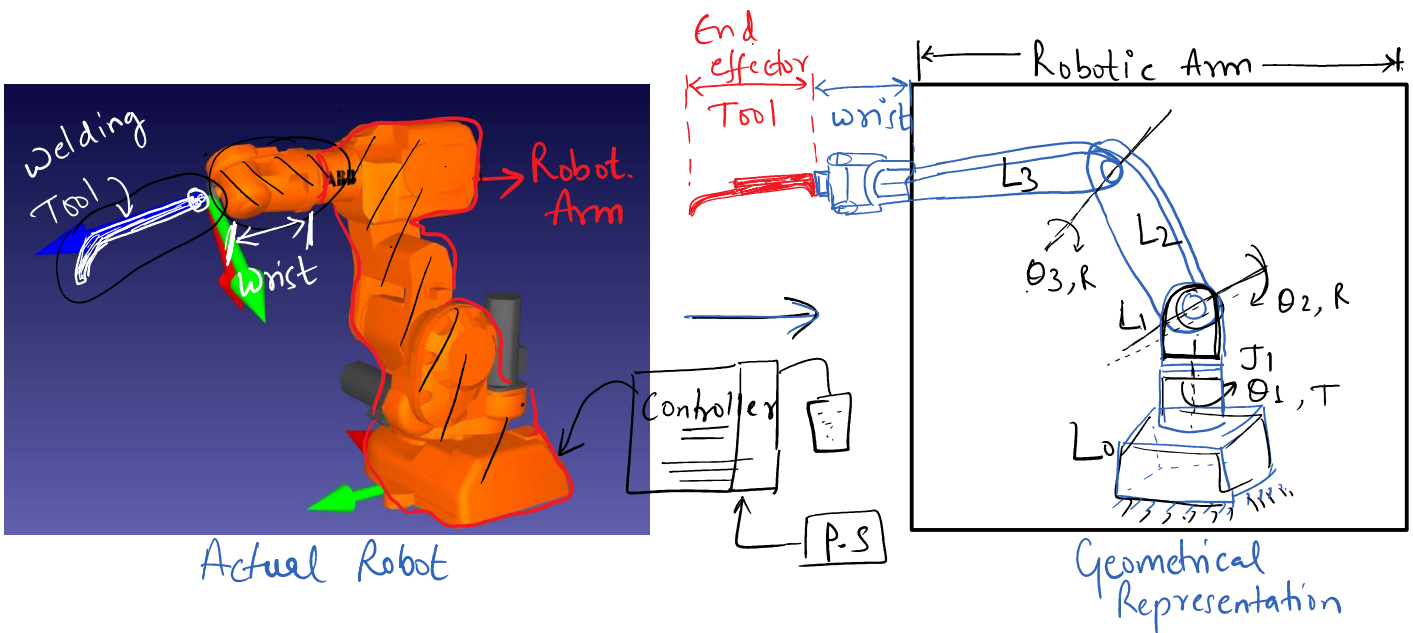
Robot Anatomy

Robot Anatomy : study of various parts and "subsystems" of Robotic system.

Study of physical construction of the robot manipulator.

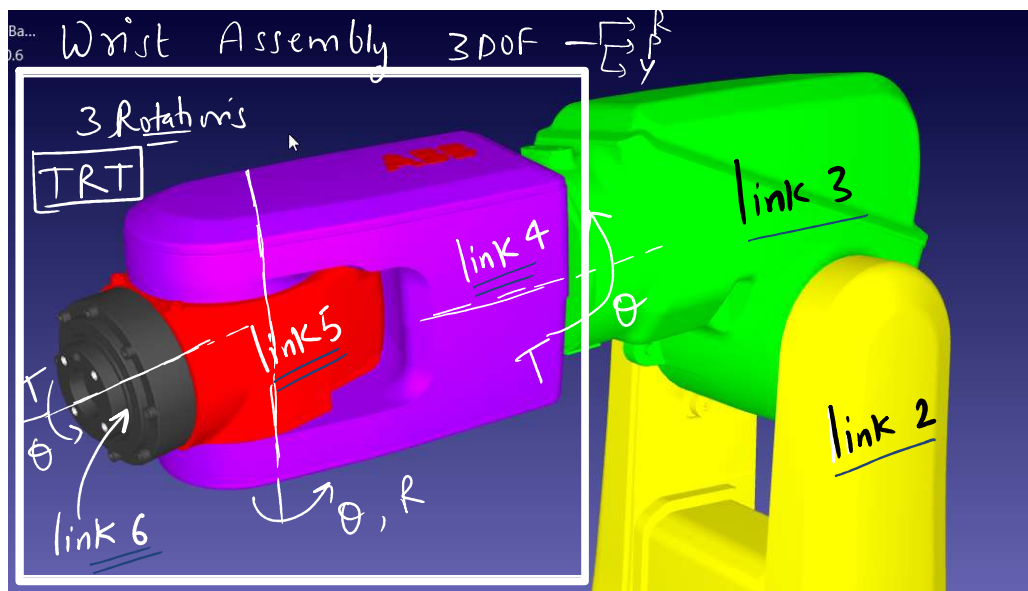
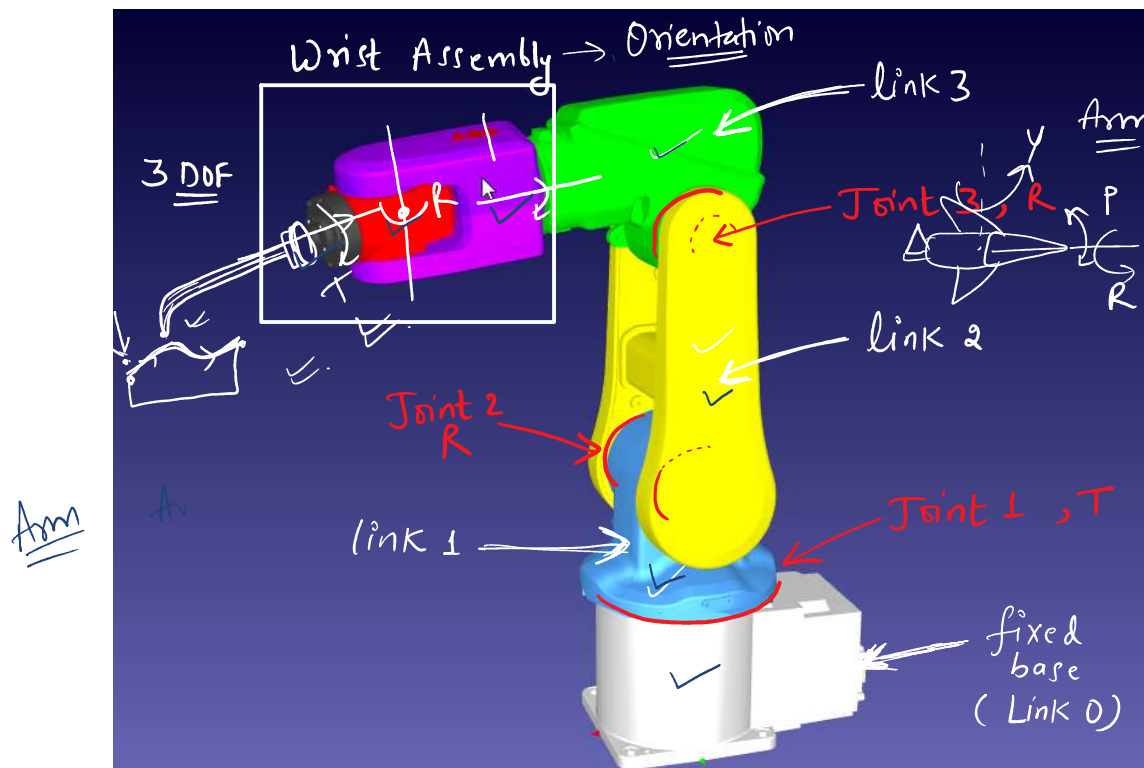
Robotic Manipulator $\Rightarrow$ Robotic Arm (3 DOF) + Robot's Wrist (3 DOF)

Actual Working Robot $\Rightarrow$ Robotic Manipulator (6 DOF) + End Effector (Tool or gripper) (May or May not have DOF)



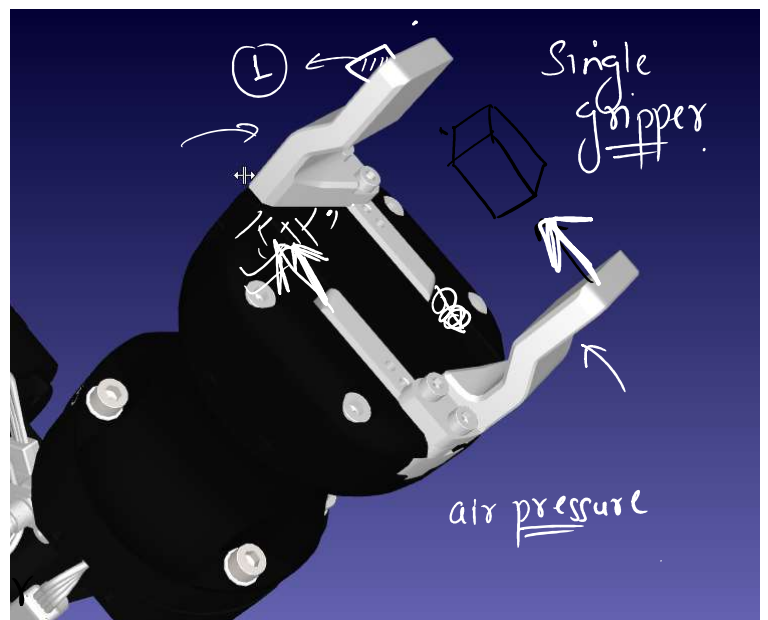
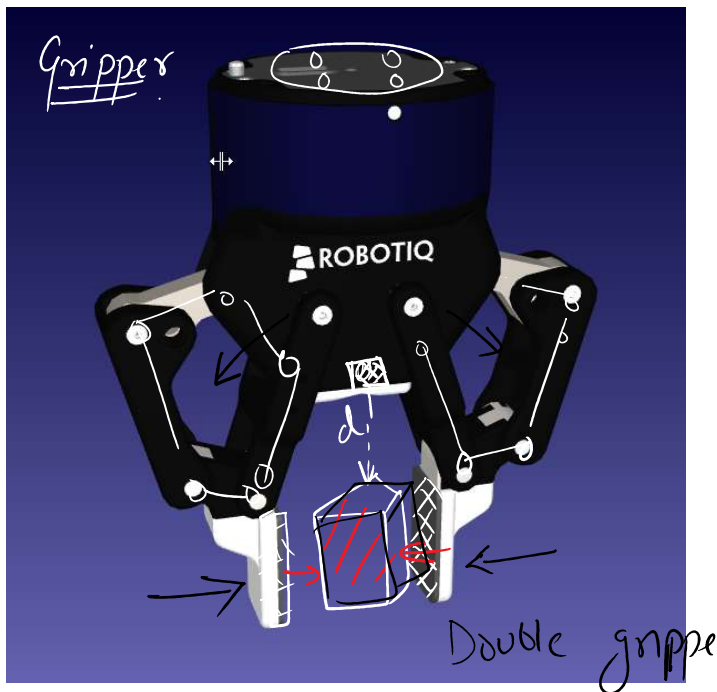
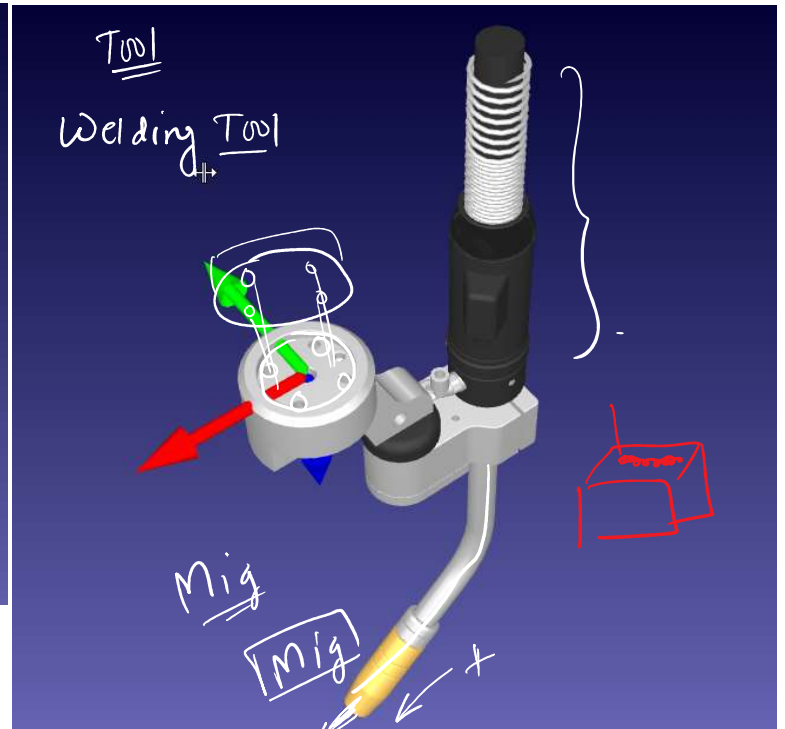
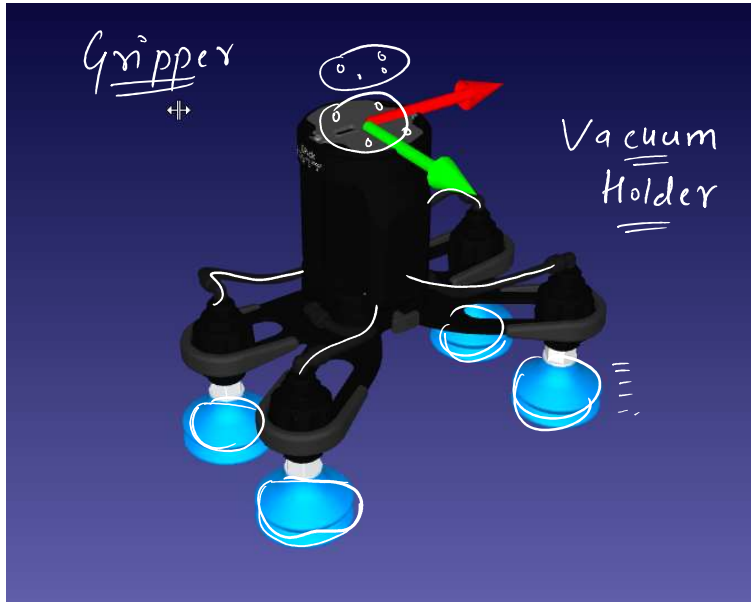
Robot's Anatomy — Robot's Wrist

- Wrist Assembly is attached to the Robot Arm's endpoint.
- Its function is to give proper orientation to the end effector.
- Wrist may have 3DOF (or lesser) but it affects dexterity.
- 3DOF (all rotations) wrist is called Roll-Pitch-Yaw wrist. [RPY wrist]
- Highest Dexterity : All three axes must intersect at a point.

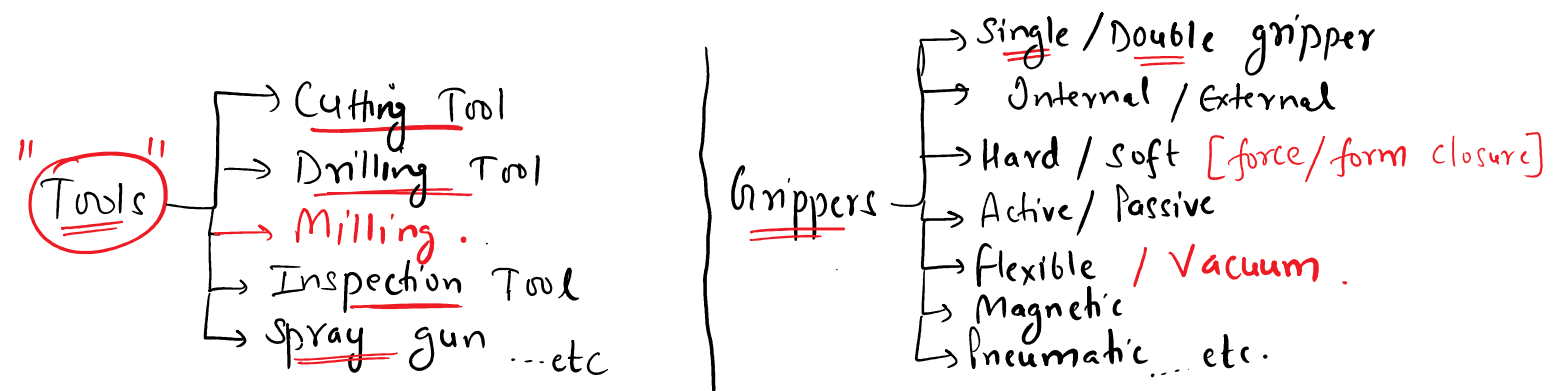


Robot's Anatomy - End Effector.

End Effector — { Tools (Directly perform work on the workpiece).
Gripper. (Grasp or hold the workpiece)



Continued

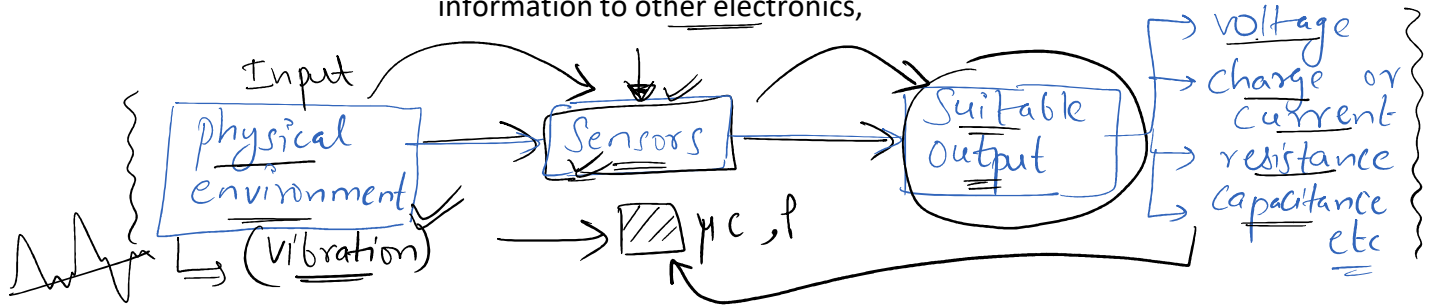


Robot's Anatomy - Sensors / Drives / Controllers

① Robotic Sensors :

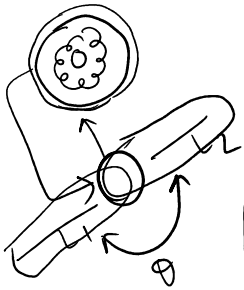
Conversion
device.

A sensor is a device that produces an output signal for the purpose of sensing a physical phenomenon. In the broadest definition, a sensor is a device, module, machine, or subsystem that detects events or changes in its environment and sends the information to other electronics,



Robotic sensors

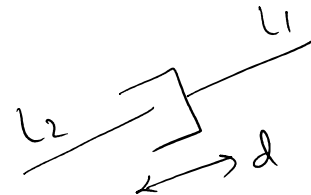
- Senses the Manipulator's joint-link parameters
- Helps in motion control
- Sensor velocity, force, torque etc
- Ensures system safety ✓
- Vision / Inspection etc.



Sensors

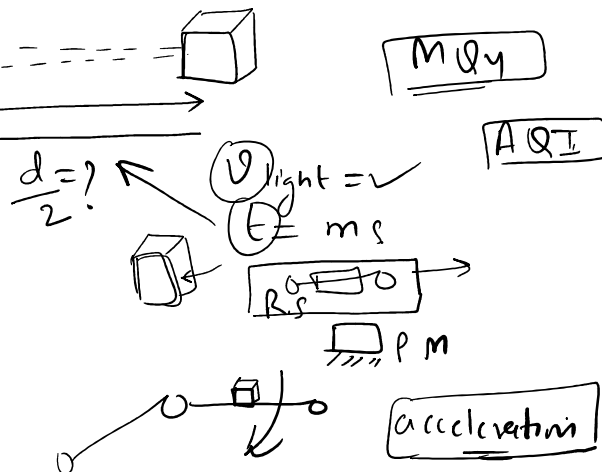
Internal Sensors

External Sensors



Some common sensors employed in Robotics

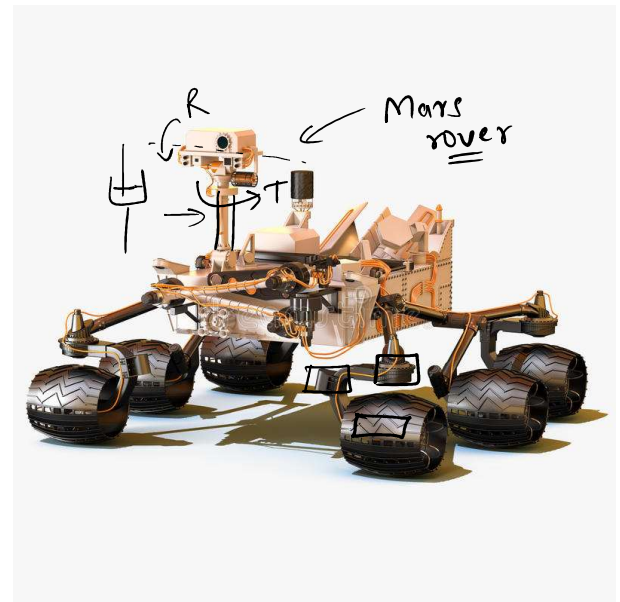
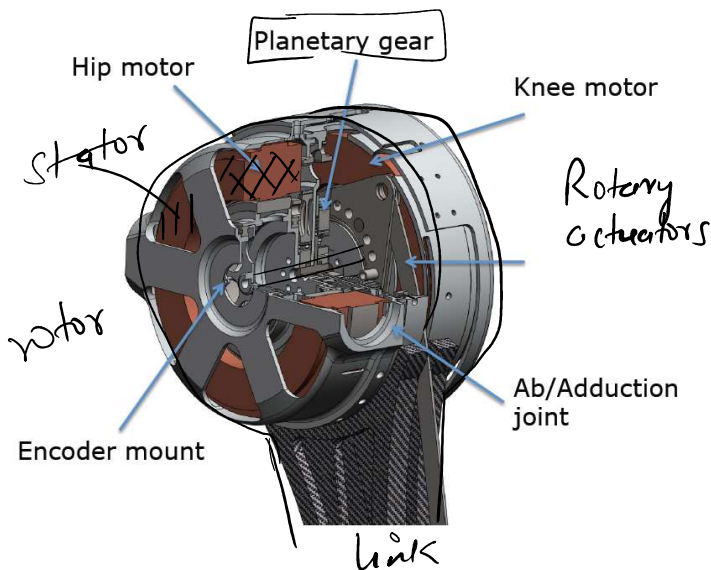
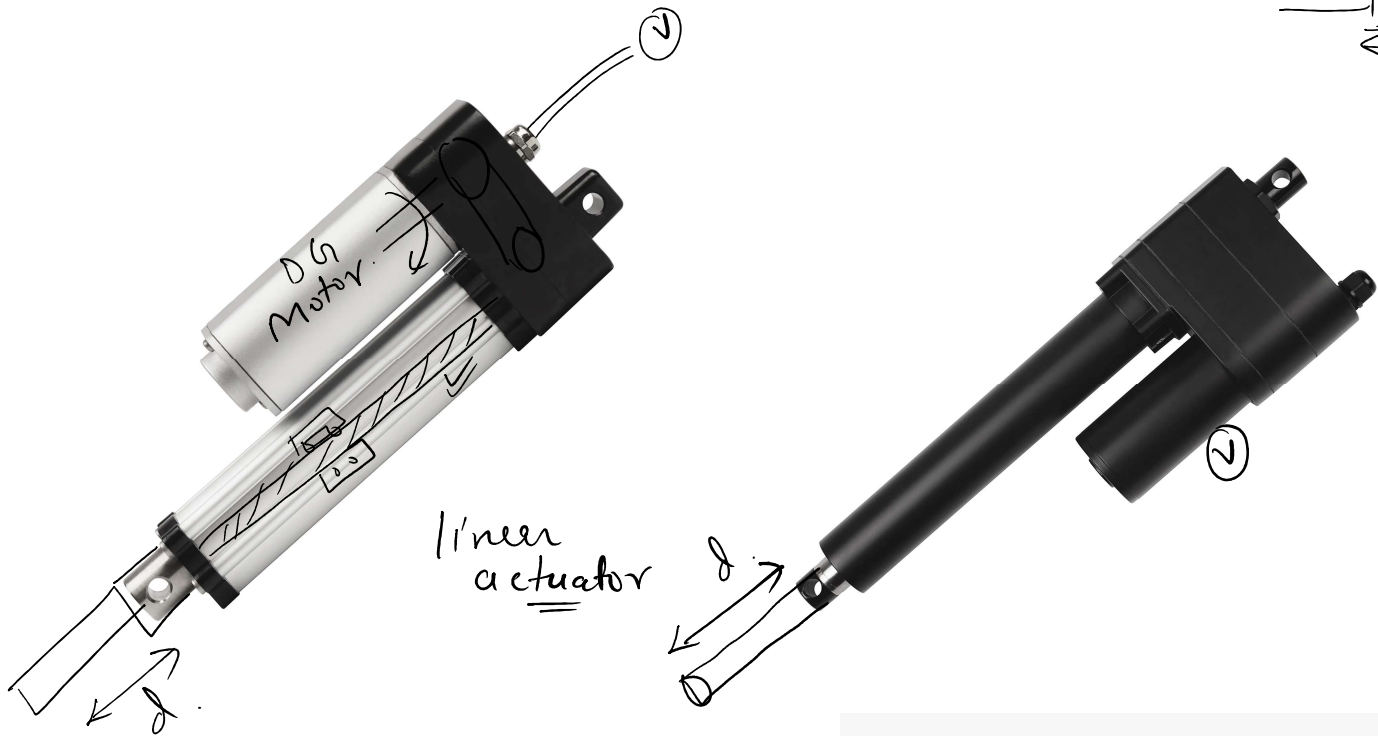
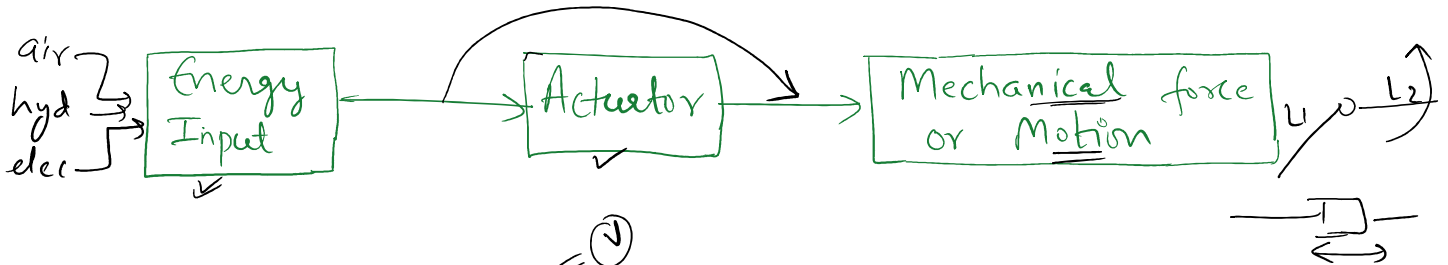
- Vision Systems (Camera)
- Range finders
- Gas Sensors ✓
- Haptic / Tactile ✓
- Limit Switches / Reed Switches
- Potentiometers and encoders
- Temperature sensors ✓
- Force / Torque Sensors ✓
- Accelerometers ✓
- Collision detection sensors ✓



Robot's Anatomy - Sensors / Drives / Controllers

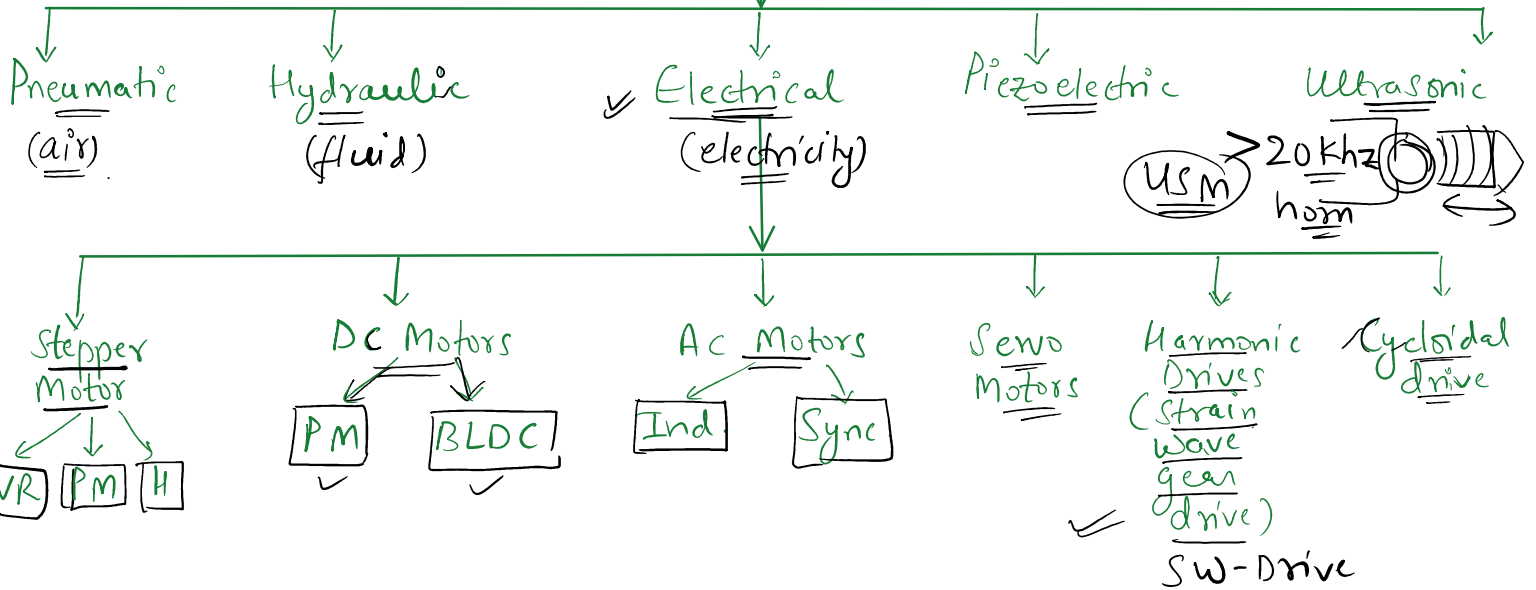
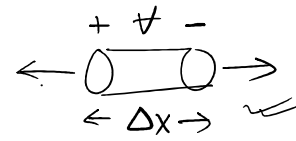
② Robot Actuators (Drives) : energy $\rightarrow$ [D] $\rightarrow$ [M.F,T]

An actuator is a part of a device or machine that helps it to achieve physical movements by converting energy, often electrical, air, or hydraulic, into mechanical force. Simply put, it is the component in any machine that enables movement.





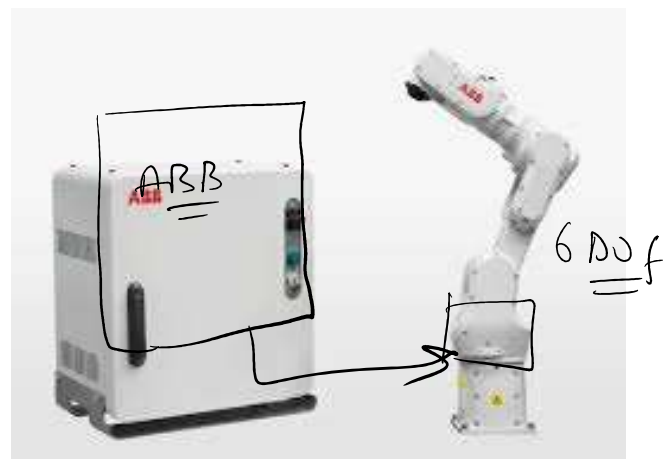
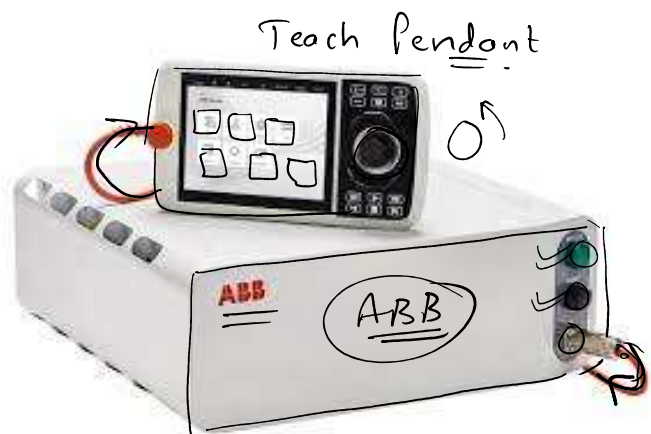
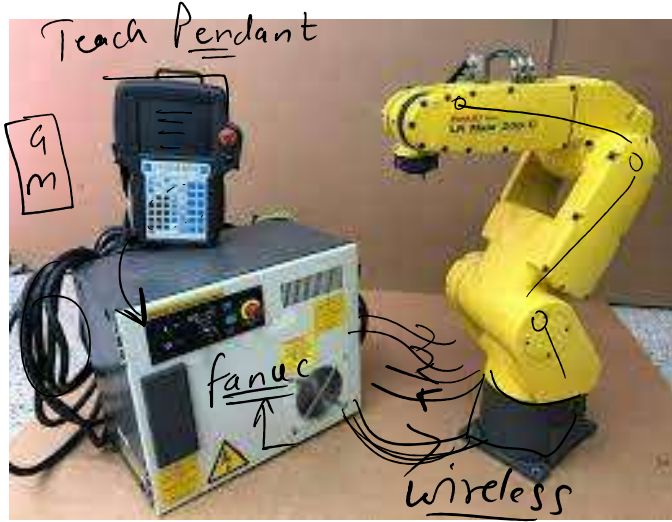
Actuator Types



Robot - Controller and Programming

Robotic control is the system that contributes to the movement of robots. This involves the mechanical aspects and programmable systems that makes it possible to control robots.

Controller → Brain of The Robotic System.



Robot Programming Methods



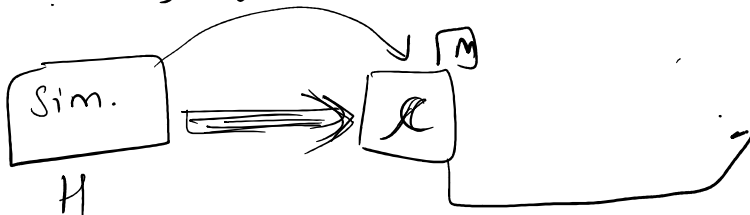
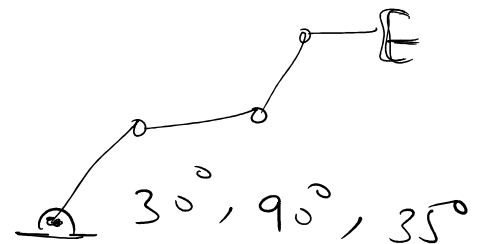
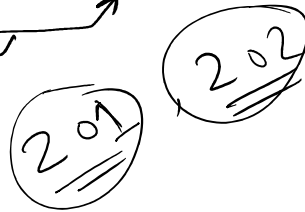
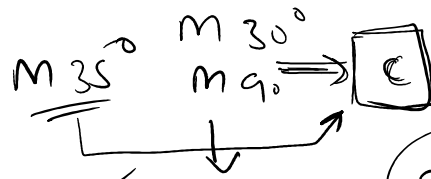
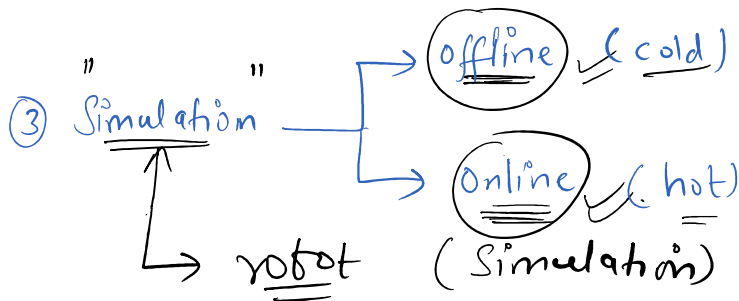
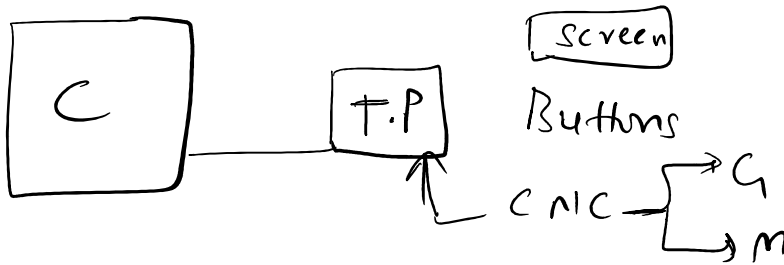
1 ~~2~~ 2

Teaching a robot to complete its workcycle is known as robot programming.

Programming Methods (3 Basic Methods)

① Teach by Demonstration / Teach by showing / Lead through programming

② Teaching Pendant / Textual Commands / Programming Languages.



→ Mathematical foundation
Coordinate Systems
Transformations.

HTM.

F, IK

Robot Classification and Specification

the Japanese Industrial Robot Association divides robots in 6 different classes:

Class 1: Manual handling devices: A device with multi degrees of freedom that is actuated by an operator.

Class 2: Fixed sequence robot: A device that performs the successive stages of a task according to a predetermined and fixed program.

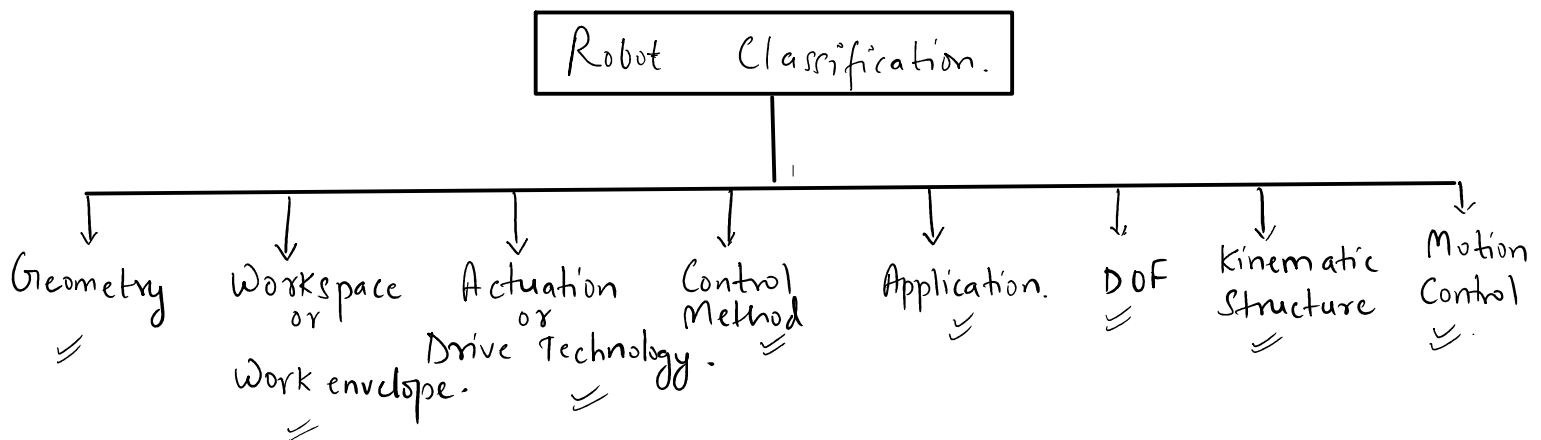
Class 3: Variable sequence robot: A device that performs the successive stages of a task according to a predetermined but programmable method.

Class 4: Playback robot: A human operator performs the task manually by leading the robot, which records the motions for later playback. The robot repeats the same motions according to the recorded information.

Class 5: Numerical control robot: The operator supplies the robot with a motion program rather than teaching it the task manually.

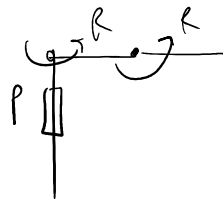
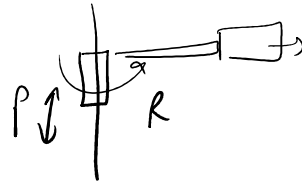
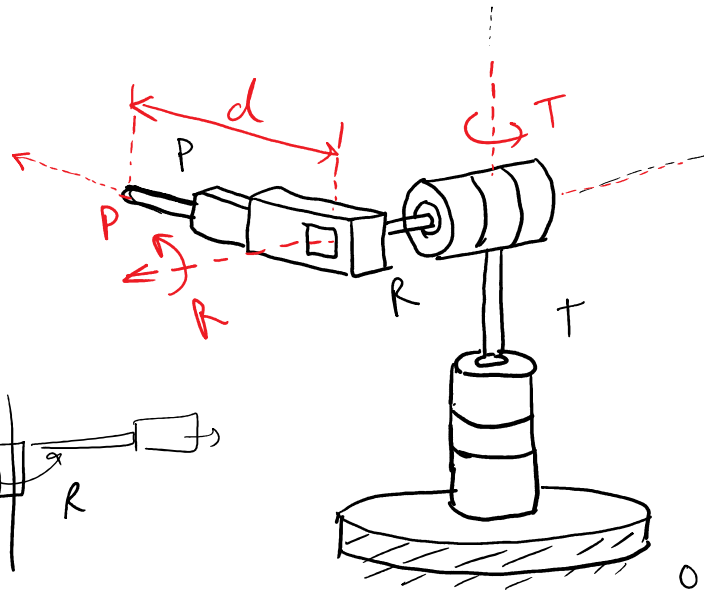
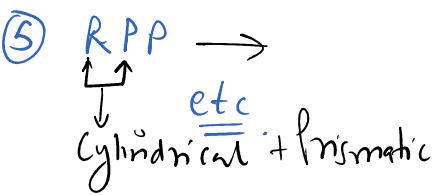
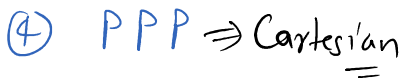
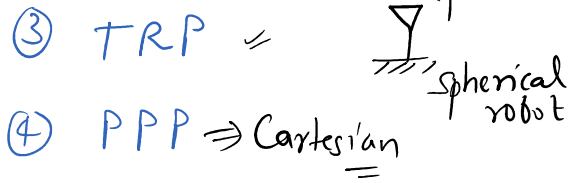
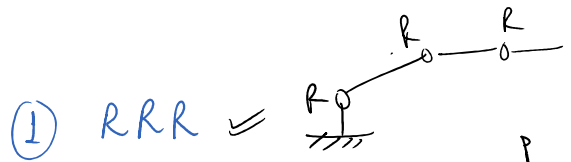
Class 6: Intelligent robot: A robot with the ability to understand its environment and the ability to successfully complete a task despite changes in the surrounding conditions under which it is to be performed.

Other than these official classifications, robots can be classified by other criteria such as geometry, workspace, actuation, control, and application.

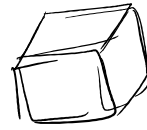


Robot Classification based on Geometry.

Based on physical and geometric differences.
Different types of joints, links and axes configurations.

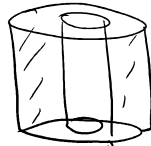


Robot Classification based on workspace or Workenvelope.



① Cartesian Coordinate Robot / Rectangular Coordinate Robot.

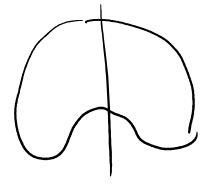
② Cylindrical Robot.



+



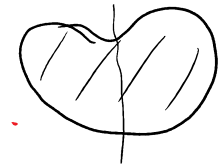
+ Spherical



③ Spherical Robot

④ Articulated / Jointed Arm Robot.

⑤ Selective compliance Assembly Robot Arm / SCARA.



Robot Classification based on Actuation or Drive Technology.

① Electrically Actuated Robots. } → Electrical Actuator } → DC Motors
Ac Motors

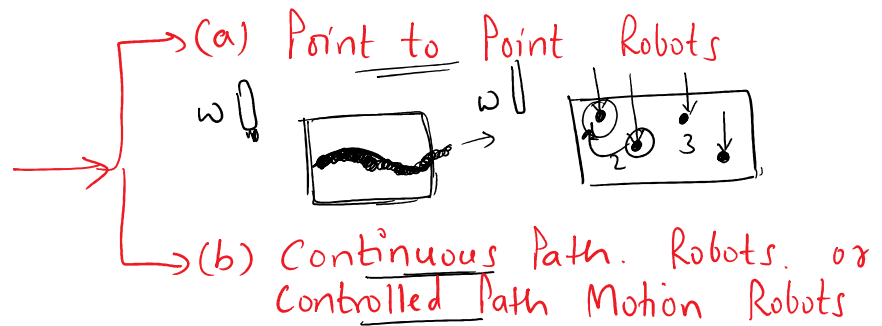
② Hydraulically Actuated Robots.] → Incompressible fluid $\boxed{P=C}$

③ Pneumatically Actuated Robots.] → Compressed air

↳ End Effector

Robot classification on the basis of Control Method.

- ① Servo Controlled
(closed loop control)
(feedback)



- ② Non Servo Controlled
(open loop control)

→ only mechanical stops are present



Point to point

Continuous path

Spot welding ✓

Pick-and-place ✓

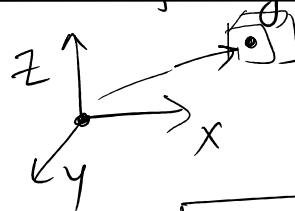
Loading and unloading ✓

Spray painting ✓

Arc welding ✓

Gluing ✓

Robot classification on the basis of Degree of freedom.



3 Dof + 3 Dof
position orientation.

- ① General Purpose Robot →

DOF = 6 { Ideal case }

- ② Redundant Robot →

DOF > 6 (Dexterity ↑)

- ③ Deficient Robot →
(under actuated robot)

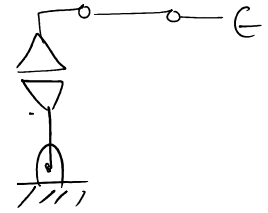
DOF < 6

Robot classification based on Kinematic Structure.

Based upon structural Topology:

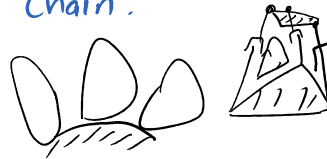
① Serial Robots or Open loop Manipulator.

- open kinematic chain



② Parallel Robots or Closed loop Manipulators.

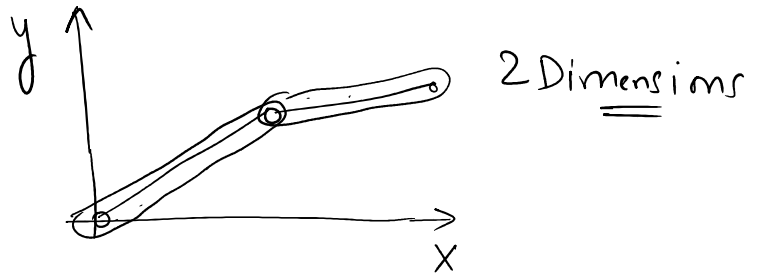
- closed kinematic chain



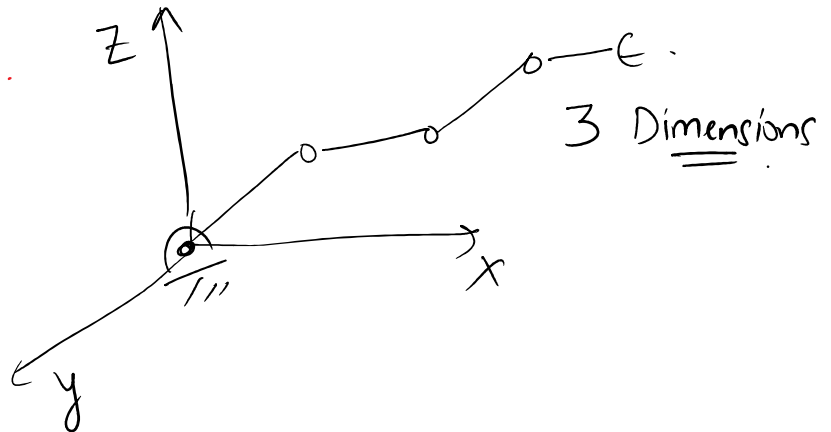
Stewart Platform

Robot classification based on Motion Characteristics

① Planar Manipulator



② Spatial Manipulator.



Robot classification on the basis of Applications.

Based on application

Assembly Robots

Non Assembly Robots

Material handling ✓

Spot welding ✓

Arc welding ✓

Spray painting and finishing ✓

Mechanical assembly ✓

Electronic assembly ✓

Material removal ✓

Inspection and testing ✓

Water jet cutting ✓

Other

Medical Robots

Military Robots

Academic/Research Robots

Under water / Aerial Robots
etc.

Robot "Specification":

While the drive technologies, work-envelope geometries, and motion control methods provide convenient ways to broadly classify robots, there are a number of additional characteristics that allow the user to further specify robotic manipulators.

Some of these robotic specifications are as follows:

Number of axes $\Leftarrow$ --- Quantity
Load Carrying Capacity/Payload --- Kg.
Maximum speed, cycle time --- mm/sec
Reach and stroke --- mm
Tool Orientation --- deg.
Repeatability --- mm
Precision and Accuracy --- mm

Operating Environment $\rightarrow \{T, W, P\}$
Resolution (programming & control) $\{D, V\}$
 $\{I, R, H\}$

Other less important parameters are.

$\rightarrow$ Brand. $\checkmark$
 $\rightarrow$ Weight $\checkmark$
 $\rightarrow$ Dof $\checkmark$
 $\rightarrow$ Robot Type $\checkmark$
etc.

